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Lecturer Name	:	VATHSALA DEVI A/P KUNALAN
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Group No.	:	21

Student ID	Student Name
TP070844	HEMANTHRAJAH A/L N KRISHNARAJAH
TP071194	ADAM JEROME JOSEPH
TP071154	ANUSHA DEVY A/P BRAHMATARAN
TP070514	HEYYMANATHAPILLAY A/L VARATHA RAJU
TP070996	SAILESS RAJ A/L RAJENDRAN

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1.0 Introduction

The goal of the Flavours Emporium food ordering system is to give both customers and vendors an easy and effective experience. By bringing together customers, vendors, delivery runners, managers and administrators on a single platform, it streamlines the ordering process. The system, which was created with Java and Object-Oriented Programming (OOP) principles, guarantees flexibility and maintainability for upcoming improvements.

Flavours Emporium's main goal is to make ordering food easy and convenient. Customers can browse menus from different vendors, add products to their carts, and request delivery, takeout, or dining in. They can also view transaction history, reorder previous foods, track order status, review order history, and provide complaints or feedback.

There are five roles in this system that are customers, vendors, delivery runners, managers and administrators. Administrators hold the highest position in this for ordering system where they are responsible for creating and managing user accounts. The design and implementation of Flavours Emporium are thoroughly explained in this study, with special attention paid to the application of Object-Oriented Programming techniques and a modular approach to system development.

2.0 Contents

2.1 Design Solutions

2.1.1 Use Case Diagram

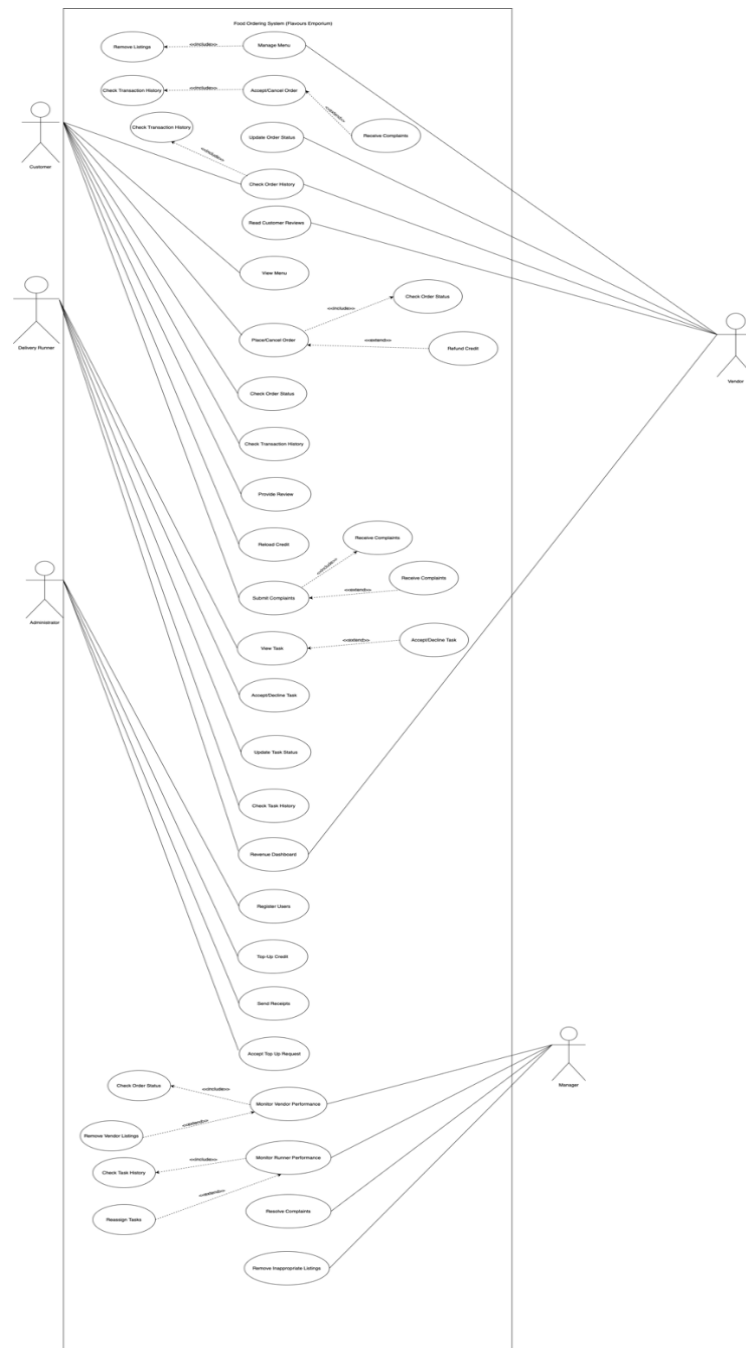


Figure 1 Flavours Emporium Use Case Diagram

(<https://drive.google.com/file/d/17pLjpOF7XwO9zTBLN9HEUp3UrfGviih/view?usp=sharing>)

2.1.2 Class Diagram

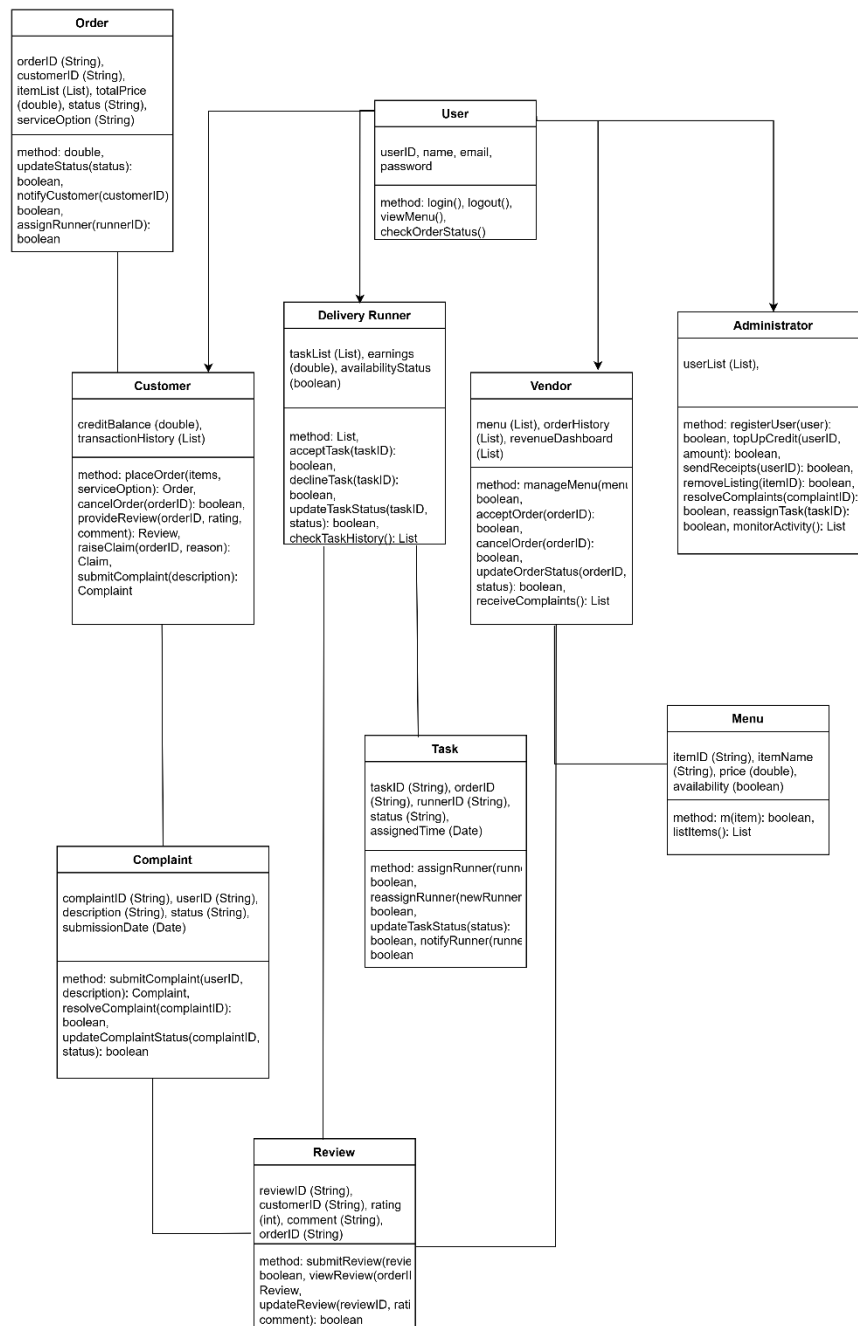


Figure 2 Flavours Emporium Class Diagram

2.2 Screenshots of Output

2.2.1 Vendor

2.2.1.1 Vendor's Login Page

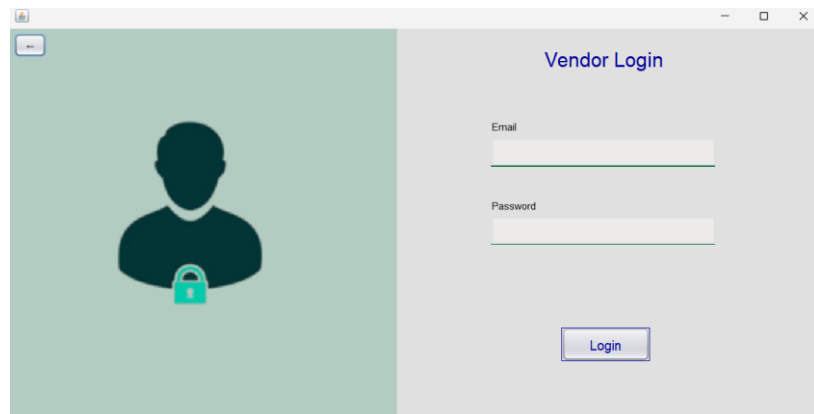


Figure 3 Vendor's Login Page

The Vendor Login will allow the vendors to access their account in a safe and secure manner by entering their registered email address and password. The interface has the input fields for both credentials, which will let a secure authentication. A login button is given to process the login request and verify the authentication details.

2.2.1.2 Vendor's Homepage

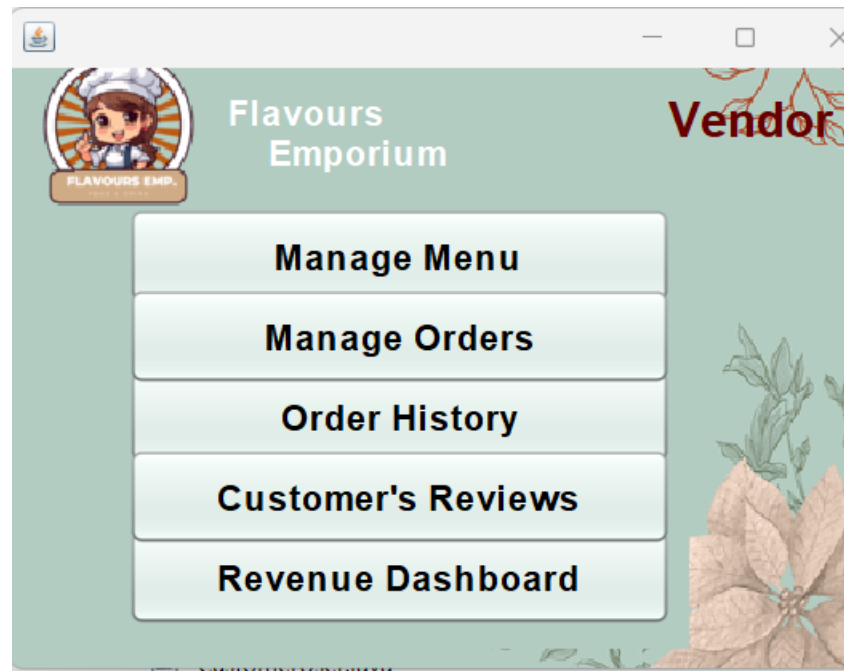
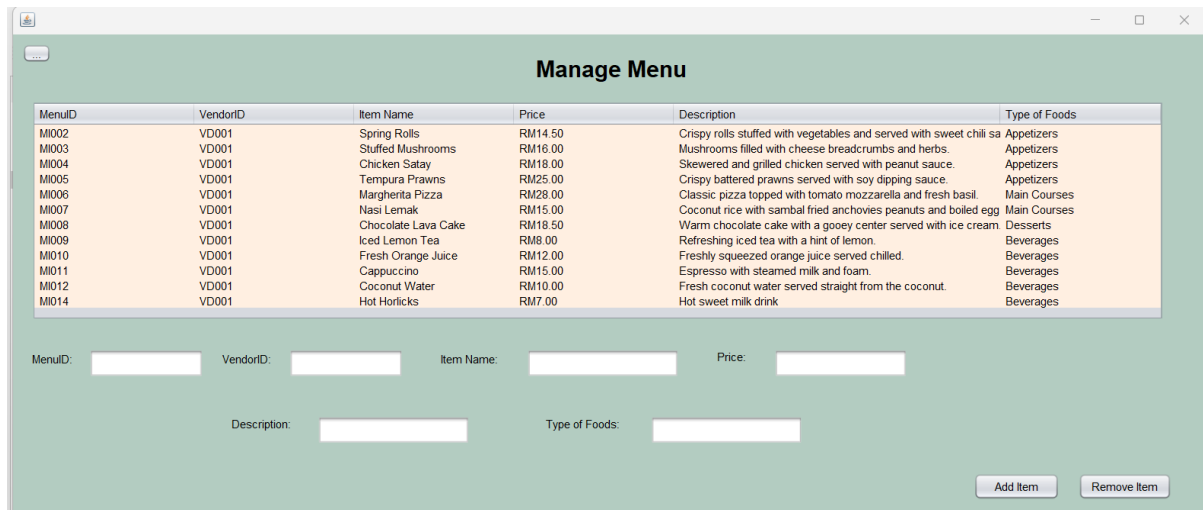


Figure 4 Vendor's Homepage

The Vendor's Homepage is a fixed interface designed to streamline restaurant management for vendors. It will provide the main functionalities such as the menu management, where the vendors can add, edit, or remove food items, and managing the orders which will enable them to track and process the incoming customer orders in a very efficient manner. The dashboard also includes an order history section for reviewing the old transactions and a customer reviews panel to view the feedback and upgrade the service quality. In addition, a revenue dashboard gives the financial insights by showing the earnings and the sales performance. With its intuitive layout, the Vendor's Homepage make sure the navigation is easy, and it will enhance the operational efficiency.

2.2.1.3 Vendor's Manage Menu Page



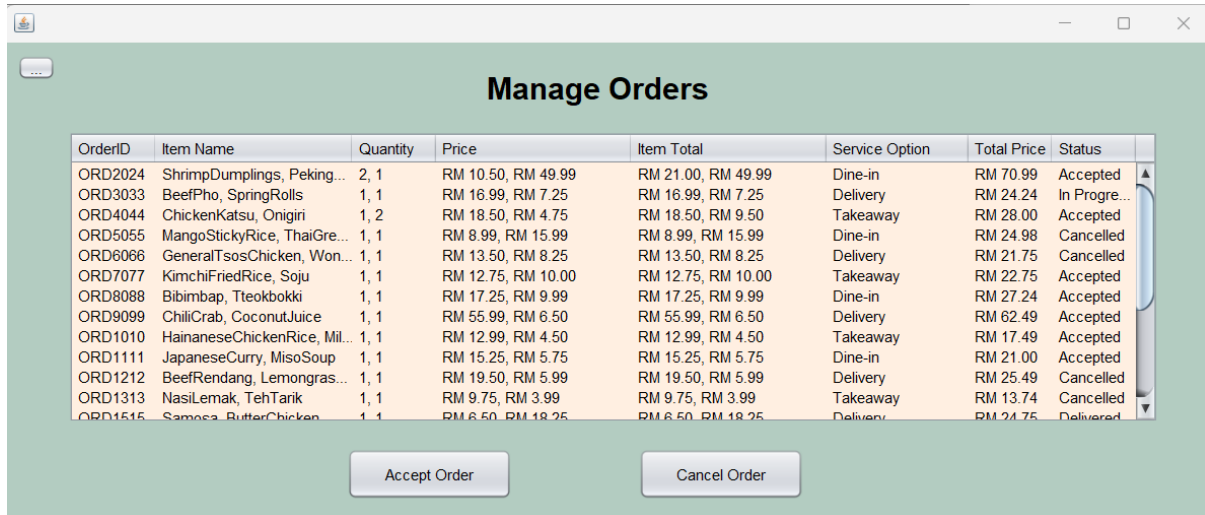
The screenshot displays a web application window titled "Manage Menu". It features a table with the following columns: MenuID, VendorID, Item Name, Price, Description, and Type of Foods. The table contains 14 rows of menu items. Below the table, there are input fields for MenuID, VendorID, Item Name, Price, Description, and Type of Foods. At the bottom right, there are two buttons: "Add Item" and "Remove Item".

MenuID	VendorID	Item Name	Price	Description	Type of Foods
MI002	VD001	Spring Rolls	RM14.50	Crispy rolls stuffed with vegetables and served with sweet chili sa	Appetizers
MI003	VD001	Stuffed Mushrooms	RM16.00	Mushrooms filled with cheese breadcrumbs and herbs.	Appetizers
MI004	VD001	Chicken Satay	RM18.00	Skewered and grilled chicken served with peanut sauce.	Appetizers
MI005	VD001	Tempura Prawns	RM25.00	Crispy battered prawns served with soy dipping sauce.	Appetizers
MI006	VD001	Margherita Pizza	RM28.00	Classic pizza topped with tomato mozzarella and fresh basil.	Main Courses
MI007	VD001	Nasi Lemak	RM15.00	Coconut rice with sambal fried anchovies peanuts and boiled egg	Main Courses
MI008	VD001	Chocolate Lava Cake	RM18.50	Warm chocolate cake with a gooey center served with ice cream.	Desserts
MI009	VD001	Iced Lemon Tea	RM8.00	Refreshing iced tea with a hint of lemon.	Beverages
MI010	VD001	Fresh Orange Juice	RM12.00	Freshly squeezed orange juice served chilled.	Beverages
MI011	VD001	Cappuccino	RM15.00	Espresso with steamed milk and foam.	Beverages
MI012	VD001	Coconut Water	RM10.00	Fresh coconut water served straight from the coconut.	Beverages
MI014	VD001	Hot Horlicks	RM7.00	Hot sweet milk drink	Beverages

Figure 5 Vendor's Manage Menu Page

The Manage Menu Page is a structured dashboard designed which will help the vendors to manage their food order efficiently with their food orderings. It will show a detailed table which contains menu items, such as MenuID, VendorID, ItemName, Price, Description, and Type of Foods. Vendors can add the menu items in a very easy way by entering relevant details in the given text fields and clicking the "Add Item" button. Similarly, they can remove the selected men item using the "Remove Items". The "Homepage" button allows users to bring them back to Vendor's Homepage. This interface will make sure a seamless menu updates, enhancing the order management and customer satisfaction.

2.2.1.4 Vendor's Manage Order Page



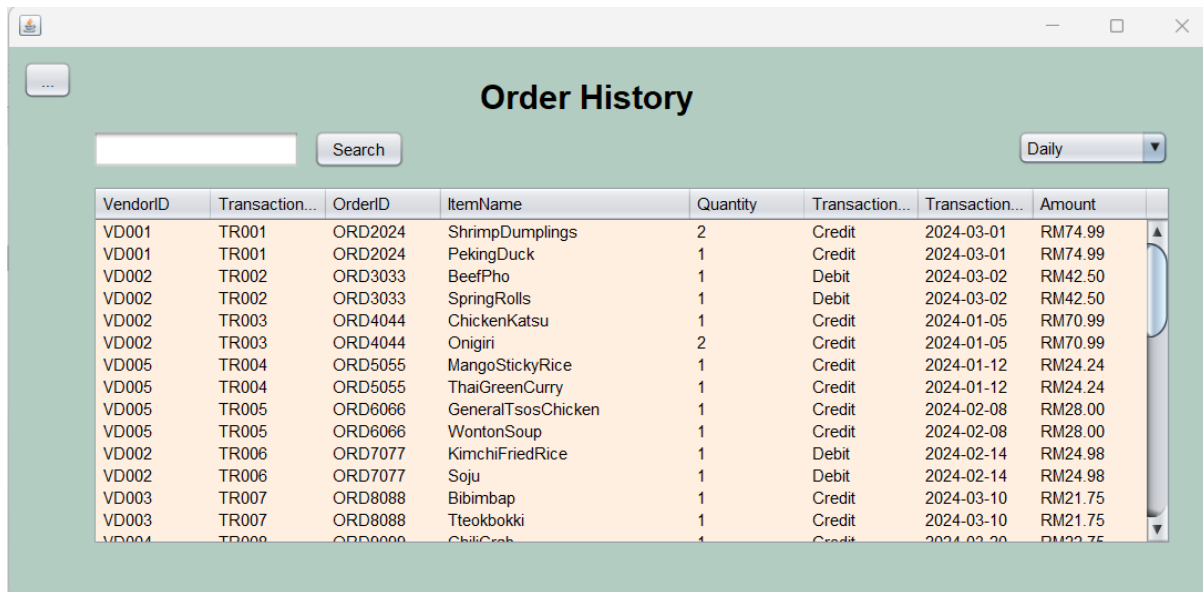
OrderID	Item Name	Quantity	Price	Item Total	Service Option	Total Price	Status
ORD2024	ShrimpDumplings, Peking...	2, 1	RM 10.50, RM 49.99	RM 21.00, RM 49.99	Dine-in	RM 70.99	Accepted
ORD3033	BeefPho, SpringRolls	1, 1	RM 16.99, RM 7.25	RM 16.99, RM 7.25	Delivery	RM 24.24	In Progre...
ORD4044	ChickenKatsu, Onigiri	1, 2	RM 18.50, RM 4.75	RM 18.50, RM 9.50	Takeaway	RM 28.00	Accepted
ORD5055	MangoStickyRice, ThaiGre...	1, 1	RM 8.99, RM 15.99	RM 8.99, RM 15.99	Dine-in	RM 24.98	Cancelled
ORD6066	GeneralTsoChicken, Won...	1, 1	RM 13.50, RM 8.25	RM 13.50, RM 8.25	Delivery	RM 21.75	Cancelled
ORD7077	KimchiFriedRice, Soju	1, 1	RM 12.75, RM 10.00	RM 12.75, RM 10.00	Takeaway	RM 22.75	Accepted
ORD8088	Bibimbap, Tteokbokki	1, 1	RM 17.25, RM 9.99	RM 17.25, RM 9.99	Dine-in	RM 27.24	Accepted
ORD9099	ChiliCrab, CoconutJuice	1, 1	RM 55.99, RM 6.50	RM 55.99, RM 6.50	Delivery	RM 62.49	Accepted
ORD1010	HainaneseChickenRice, Mil...	1, 1	RM 12.99, RM 4.50	RM 12.99, RM 4.50	Takeaway	RM 17.49	Accepted
ORD1111	JapaneseCurry, MisoSoup	1, 1	RM 15.25, RM 5.75	RM 15.25, RM 5.75	Dine-in	RM 21.00	Accepted
ORD1212	BeefRendang, Lemongras...	1, 1	RM 19.50, RM 5.99	RM 19.50, RM 5.99	Delivery	RM 25.49	Cancelled
ORD1313	NasiLemak, TehTarik	1, 1	RM 9.75, RM 3.99	RM 9.75, RM 3.99	Takeaway	RM 13.74	Cancelled
ORD1515	Samosa, ButterChicken	1, 1	RM 6.50, RM 18.25	RM 6.50, RM 18.25	Delivery	RM 24.75	Delivered

Accept Order Cancel Order

Figure 6 Vendor's Manage Order Page

The Manage Order Page is a well-structured dashboard is to be designed to update order processing for vendors by showing the key details such as the OrderID, ItemName, Quantity, Price, Item Total, Service Option, Total Price, and Status. With this layout, vendors can efficiently manage incoming orders by accepting or cancelling them using the integrated buttons. The “Accept Order” button will allow the vendors to approve the pending orders for fulfilment, while the “Cancel Order” button will give the option to cancel the orders when needed.

2.2.1.5 Vendor's Order History Page

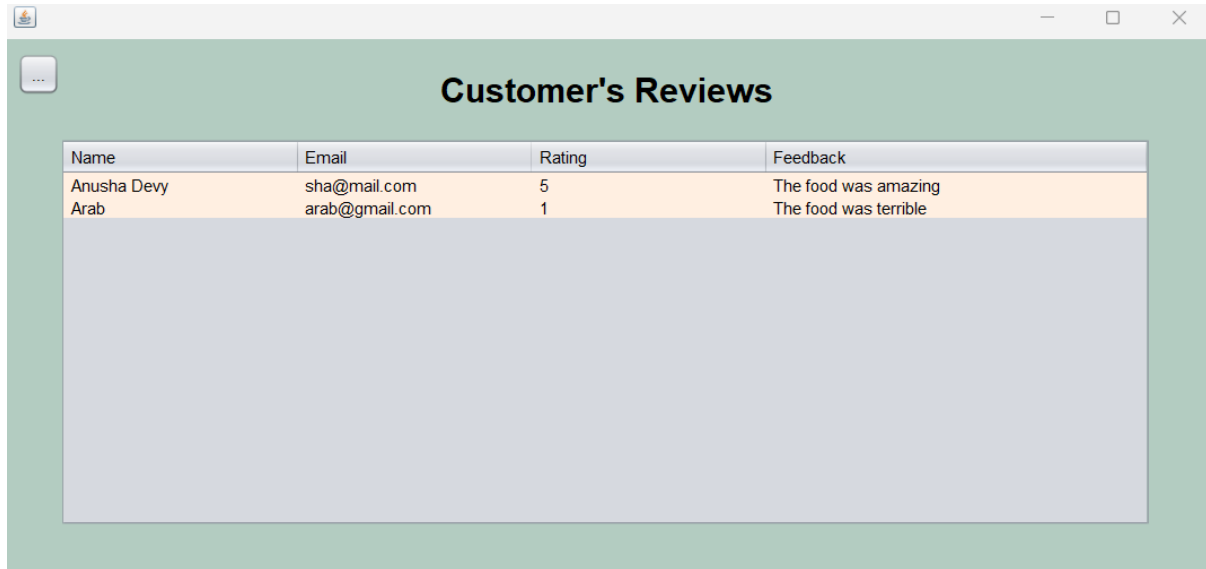


VendorID	Transaction...	OrderID	ItemName	Quantity	Transaction...	Transaction...	Amount
VD001	TR001	ORD2024	ShrimpDumplings	2	Credit	2024-03-01	RM74.99
VD001	TR001	ORD2024	PekingDuck	1	Credit	2024-03-01	RM74.99
VD002	TR002	ORD3033	BeefPho	1	Debit	2024-03-02	RM42.50
VD002	TR002	ORD3033	SpringRolls	1	Debit	2024-03-02	RM42.50
VD002	TR003	ORD4044	ChickenKatsu	1	Credit	2024-01-05	RM70.99
VD002	TR003	ORD4044	Onigiri	2	Credit	2024-01-05	RM70.99
VD005	TR004	ORD5055	MangoStickyRice	1	Credit	2024-01-12	RM24.24
VD005	TR004	ORD5055	ThaiGreenCurry	1	Credit	2024-01-12	RM24.24
VD005	TR005	ORD6066	GeneralTsosChicken	1	Credit	2024-02-08	RM28.00
VD005	TR005	ORD6066	WontonSoup	1	Credit	2024-02-08	RM28.00
VD002	TR006	ORD7077	KimchiFriedRice	1	Debit	2024-02-14	RM24.98
VD002	TR006	ORD7077	Soju	1	Debit	2024-02-14	RM24.98
VD003	TR007	ORD8088	Bibimbap	1	Credit	2024-03-10	RM21.75
VD003	TR007	ORD8088	Tteokbokki	1	Credit	2024-03-10	RM21.75
VD004	TR008	ORD9099	ChiliCrab	1	Credit	2024-03-20	RM32.75

Figure 7 Vendor's Order History Page

The Order History interface will provide the vendors with a record of past transactions, which will allow them to track and review the completed orders. The table will display the important details such as VendorID, TransactionID, OrderID, ItemName, Quantity, Transaction Type, Transaction Date and Amount which will make sure the clear financial and inventory tracking. The search bar at the top will let the Vendors to find the specific transactions using the VendorID. The dropdown menu suggests for filtering the transaction based on different time frames. This well-organized system will help the vendors efficiently to monitor the sales, manage the revenue and maintain a very accurate transaction records.

2.2.1.6 Vendor's Customer's Review Page



Name	Email	Rating	Feedback
Anusha Devy	sha@mail.com	5	The food was amazing
Arab	arab@gmail.com	1	The food was terrible

Figure 8 Vendor's Customer's Review Page

The Customer's Reviews gives a synopsis of feedback from customers about their food experience. This table will show the main details such as the Name, Email, Rating and Feedback, which will allow the vendors to access the customer satisfaction and will find the areas for improvements. In the picture above, gave a 5-star rating with positive feedback, while another one gave a 1-star rating with negative feedback. This system allows vendor to see the customer opinions and make the necessary adjustment to improve their service.

2.2.1.7 Vendor's Revenue Dashboard Page

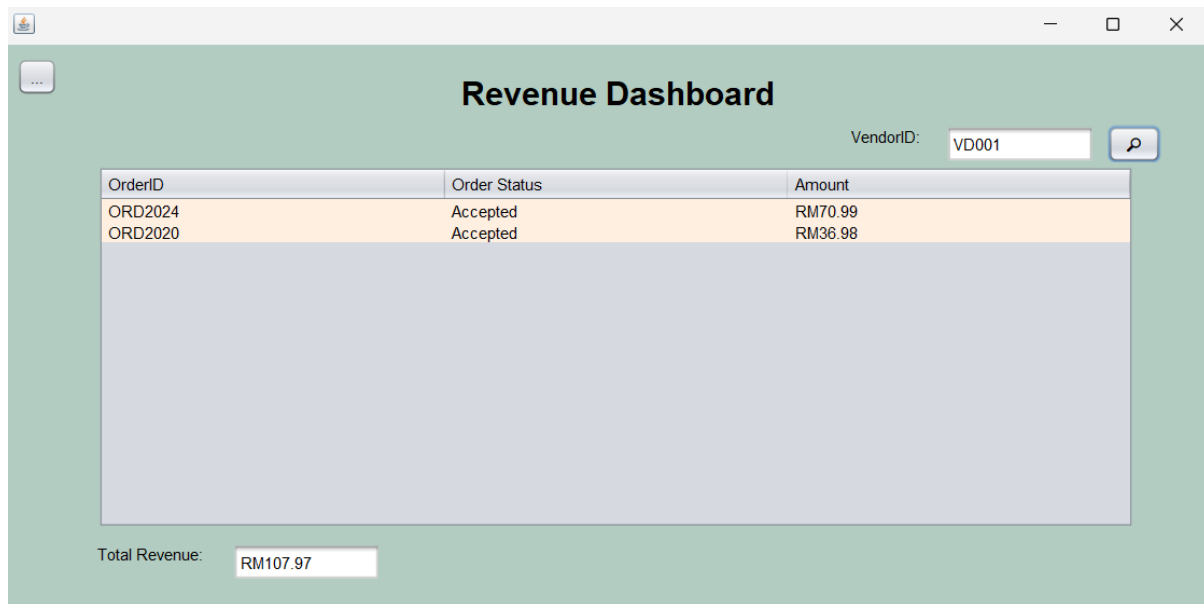


Figure 9 Vendor's Revenue Dashboard Page

The Revenue Dashboard gives an overview of a vendor's earnings by showing the accepted orders and their respective amounts. The interface has the search bar where the VendorID can be typed out to filter the revenue data for a particular vendor. Below the search bar, the table lists OrderID, Order Status and Amount which will show all the accepted orders.

2.2.2 Customer

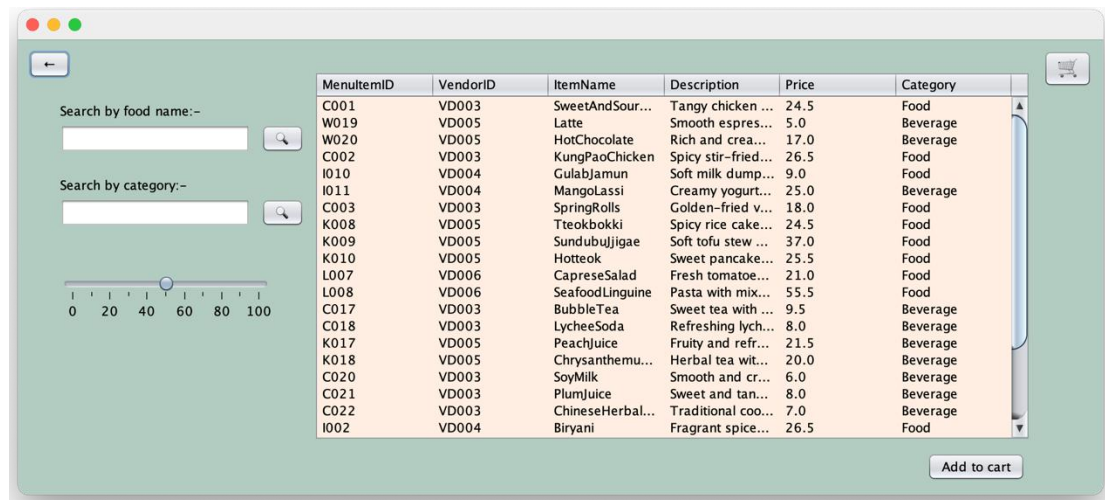


Figure 10: Menu Display Screen

This screen showcases a list of menu items available for selection. It features columns for ItemID, VendorID, ItemName, Description, Price, and Category, enabling users to view detailed information about each product. The interface also includes search options for filtering by food name and category, as well as a slider to adjust price range. The “Add to Cart” button at the bottom allows users to select and purchase items directly from the menu. This organized layout ensures a user-friendly shopping experience by offering multiple ways to browse the menu.

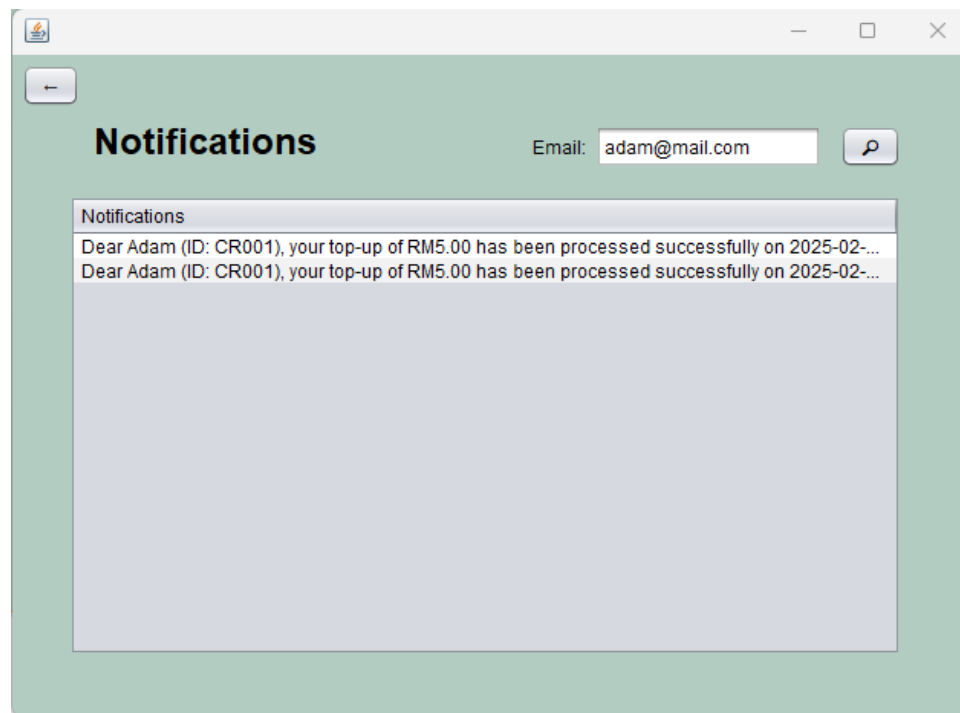
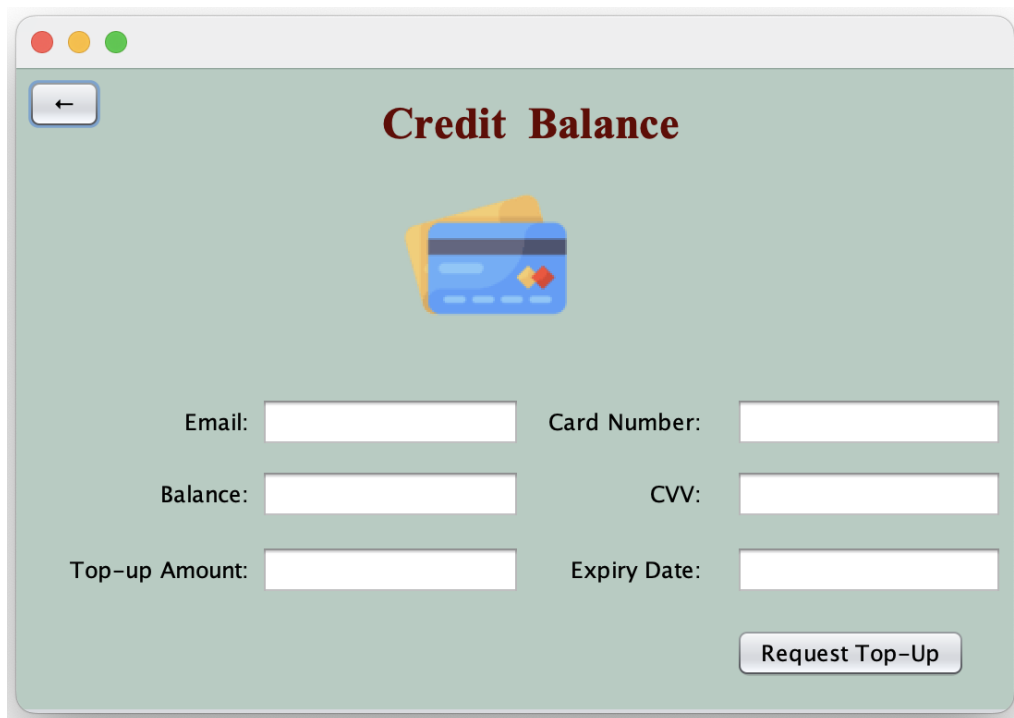


Figure 11 Notifications Screen

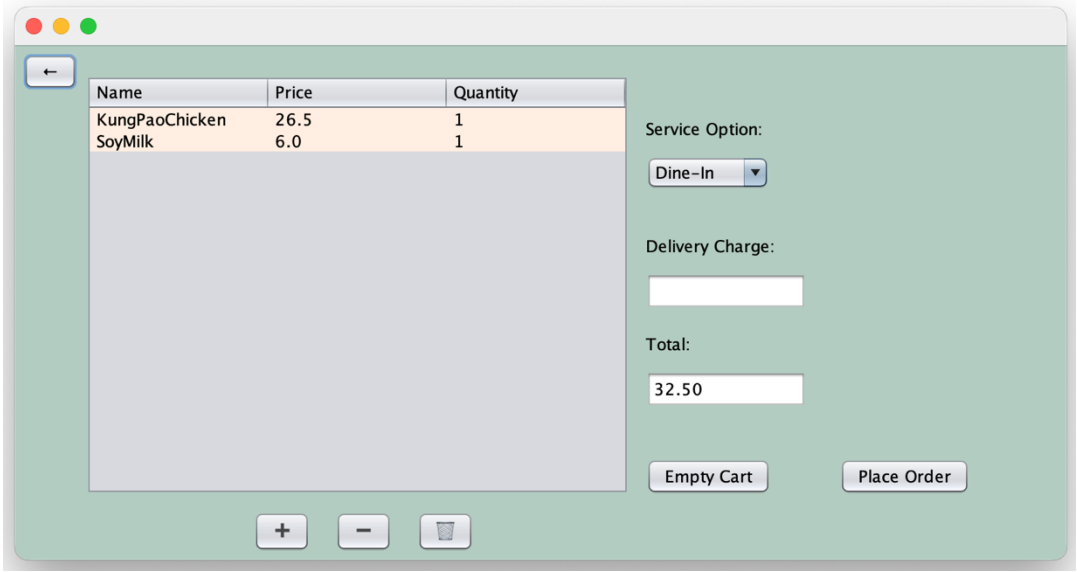
This screen is designed to display notifications relevant to the user, such as top-up notification. It features an email input field and a search button to filter notifications by email. The notification messages appear in a table format below. This setup keeps users informed about top-ups for smoother food ordering process.

A screenshot of a web application window titled "Credit Balance". The window has a light green background and a grey border. At the top left, there are three colored circles (red, yellow, green) and a back button with a left arrow. In the center, there is a graphic of a blue credit card with a yellow envelope behind it. Below the graphic, there are six input fields arranged in two columns. The left column contains "Email:", "Balance:", and "Top-up Amount:". The right column contains "Card Number:", "CVV:", and "Expiry Date:". At the bottom right, there is a button labeled "Request Top-Up".

Email:		Card Number:	
Balance:		CVV:	
Top-up Amount:		Expiry Date:	
Request Top-Up			

Figure 12: Credit Balance Screen

This interface facilitates managing and topping up the user's credit balance. It includes fields for entering card number, top-up amount, CVV, and expiry date. The "Request Top-Up" button initiates the balance update process. The page is neatly organized to ensure ease of use and security, ensuring that users can easily manage their account balances while maintaining their payment information's confidentiality.



The image shows a shopping cart interface. On the left, a table lists items in the cart. Below the table are three buttons: a plus sign, a minus sign, and a trash can icon. On the right, there are labels for 'Service Option:', 'Delivery Charge:', and 'Total:'. The 'Service Option' is a dropdown menu currently showing 'Dine-In'. The 'Delivery Charge' is an empty text box. The 'Total' is a text box showing '32.50'. At the bottom right are two buttons: 'Empty Cart' and 'Place Order'.

Name	Price	Quantity
KungPaoChicken	26.5	1
SoyMilk	6.0	1

Service Option: Dine-In

Delivery Charge:

Total: 32.50

Empty Cart Place Order

Figure 13: Shopping Cart Screen

This screen displays items added to the cart, along with their names, prices, and quantities. Users can adjust the quantity or remove items before placing an order. Additional features include selecting a service option like Dine-In, Takeaway or Delivery. If customer chooses Delivery option there will be man additional charge of 5.00. The “Empty Cart” and “Place Order” buttons provide clear call-to-action options, enhancing the shopping experience with straightforward controls for order management.

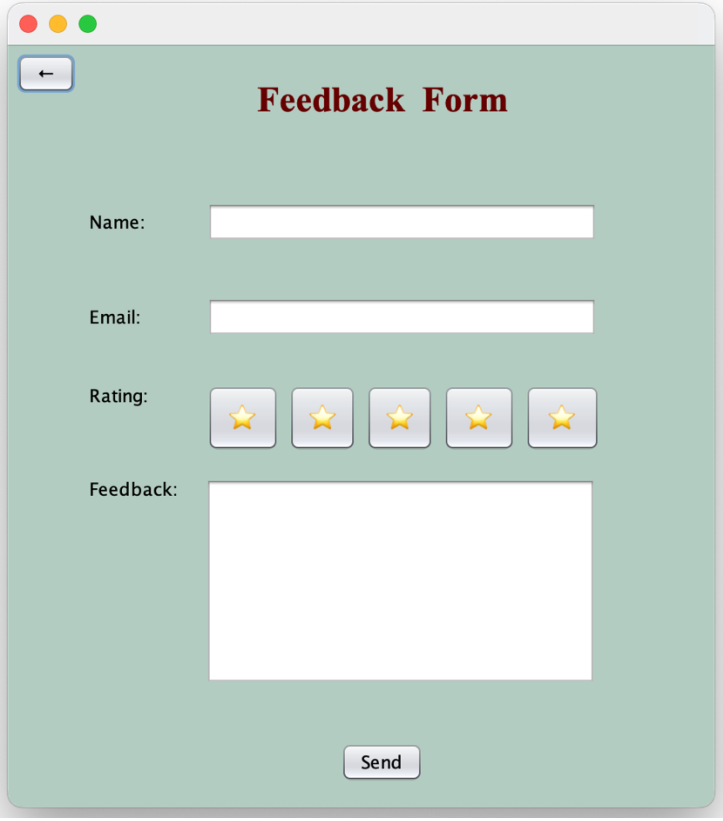
A screenshot of a web application window titled "Feedback Form". The window has a light green background and a standard macOS-style title bar with red, yellow, and green buttons. In the top-left corner of the form area is a small button with a left-pointing arrow. The form contains four input fields: "Name:" with a single-line text box, "Email:" with a single-line text box, "Rating:" with five star icons in individual boxes, and "Feedback:" with a large multi-line text area. At the bottom center is a "Send" button.

Figure 14: Feedback Form Screen

This form collects customer feedback by allowing users to input their name, email, rating, and comments. It includes star icons for rating, enabling users to express their satisfaction levels visually. The “Send” button at the bottom submits the feedback. This encourages users to share their experiences, which helps businesses gather valuable insights for improvement.

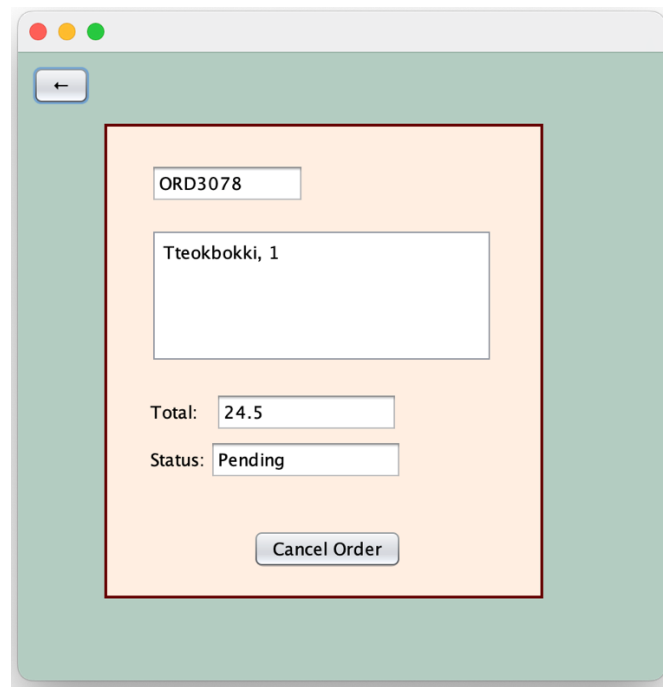
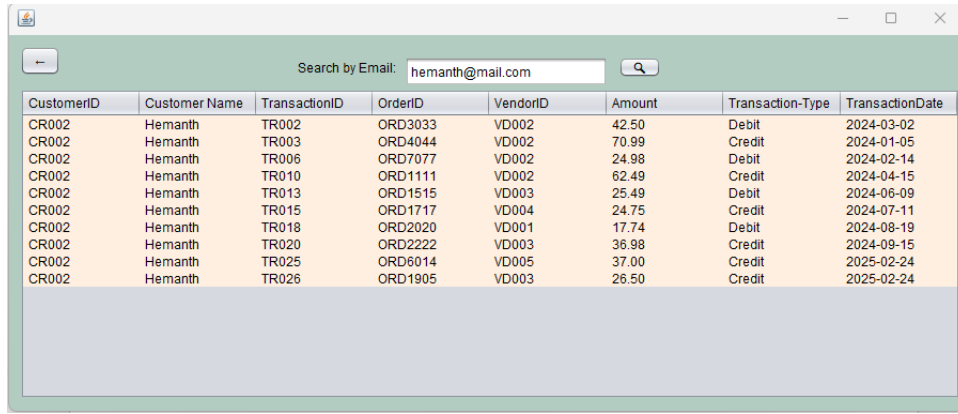
A UI mockup of an 'Order Status Screen' presented as a window with a light green background and a dark green border. In the top-left corner of the window is a small blue button with a white left-pointing arrow. The main content area is a light orange rectangle. Inside this rectangle, at the top, is a white text input field containing 'ORD3078'. Below it is a larger white text area containing 'Tteokbokki, 1'. Further down, the text 'Total:' is followed by a white text input field containing '24.5'. Below that, the text 'Status:' is followed by a white text input field containing 'Pending'. At the bottom of the orange rectangle is a rounded rectangular button with a grey gradient and the text 'Cancel Order' in black.

Figure 15: Order Status Screen

This interface displays order details, including the order ID, item names, total cost, and current status. The design provides a straightforward way for users to track their orders in real-time. The “Cancel Order” button allows users to cancel their orders if needed, offering greater control over their purchasing decisions.

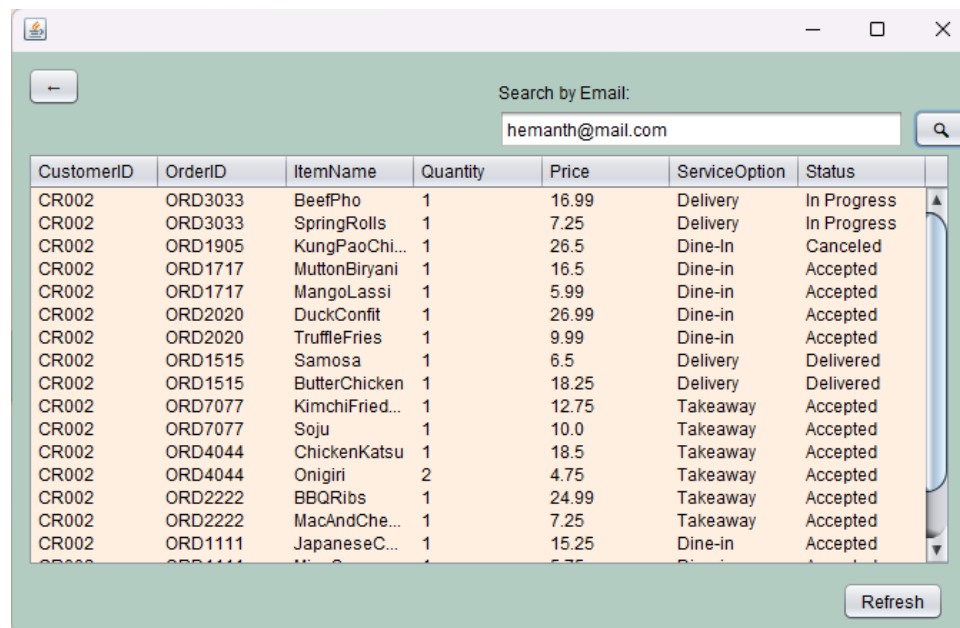


The screenshot displays a web application window titled "Transaction History Screen". At the top, there is a search bar labeled "Search by Email:" with the text "hemanth@mail.com" entered. Below the search bar is a table with 8 columns: CustomerID, Customer Name, TransactionID, OrderID, VendorID, Amount, Transaction-Type, and TransactionDate. The table contains 14 rows of transaction data for customer Hemanth. The transactions are listed in chronological order from top to bottom, starting with a debit on 2024-03-02 and ending with a credit on 2025-02-24. The table is styled with alternating light orange and light gray rows. Below the table, there is a large, empty light gray rectangular area.

CustomerID	Customer Name	TransactionID	OrderID	VendorID	Amount	Transaction-Type	TransactionDate
CR002	Hemanth	TR002	ORD3033	VD002	42.50	Debit	2024-03-02
CR002	Hemanth	TR003	ORD4044	VD002	70.99	Credit	2024-01-05
CR002	Hemanth	TR006	ORD7077	VD002	24.98	Debit	2024-02-14
CR002	Hemanth	TR010	ORD1111	VD002	62.49	Credit	2024-04-15
CR002	Hemanth	TR013	ORD1515	VD003	25.49	Debit	2024-06-09
CR002	Hemanth	TR015	ORD1717	VD004	24.75	Credit	2024-07-11
CR002	Hemanth	TR018	ORD2020	VD001	17.74	Debit	2024-08-19
CR002	Hemanth	TR020	ORD2222	VD003	36.98	Credit	2024-09-15
CR002	Hemanth	TR025	ORD6014	VD005	37.00	Credit	2025-02-24
CR002	Hemanth	TR026	ORD1905	VD003	26.50	Credit	2025-02-24

Figure 16: Transaction History Screen

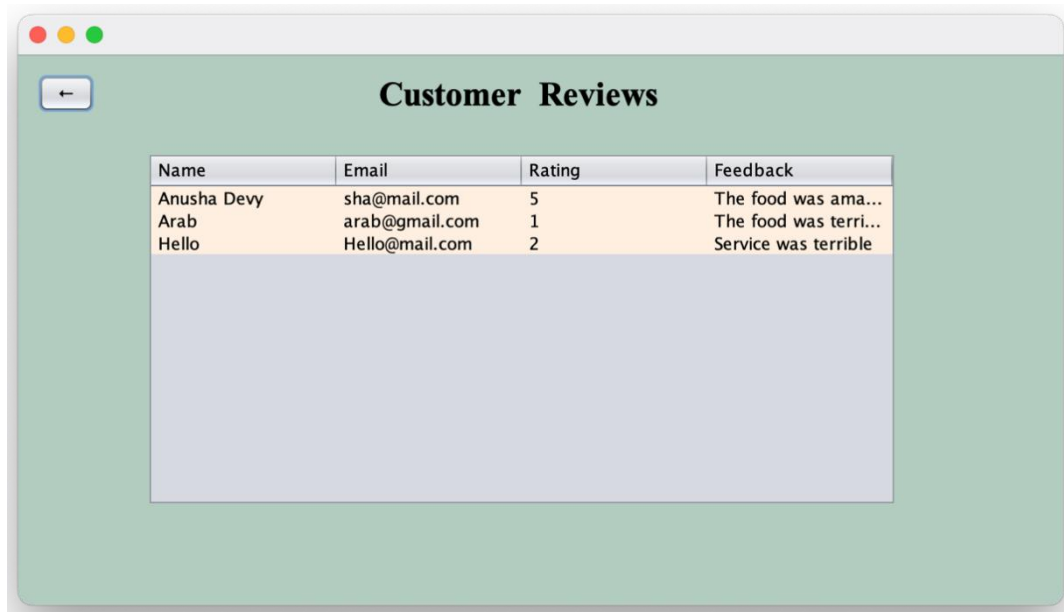
This screen showcases a comprehensive list of transactions, including CustomerID, CustomerName, TransactionID, OrderID, VendorID, Amount, Transaction Type, and Transaction Date. Users can search for transactions by email using the input field at the top. This feature helps users keep track of their purchases and spending history, promoting transparency and better financial management.



CustomerID	OrderID	ItemName	Quantity	Price	ServiceOption	Status
CR002	ORD3033	BeefPho	1	16.99	Delivery	In Progress
CR002	ORD3033	SpringRolls	1	7.25	Delivery	In Progress
CR002	ORD1905	KungPaoChi...	1	26.5	Dine-In	Canceled
CR002	ORD1717	MuttonBiryani	1	16.5	Dine-in	Accepted
CR002	ORD1717	MangoLassi	1	5.99	Dine-in	Accepted
CR002	ORD2020	DuckConfit	1	26.99	Dine-in	Accepted
CR002	ORD2020	TruffleFries	1	9.99	Dine-in	Accepted
CR002	ORD1515	Samosa	1	6.5	Delivery	Delivered
CR002	ORD1515	ButterChicken	1	18.25	Delivery	Delivered
CR002	ORD7077	KimchiFried...	1	12.75	Takeaway	Accepted
CR002	ORD7077	Soju	1	10.0	Takeaway	Accepted
CR002	ORD4044	ChickenKatsu	1	18.5	Takeaway	Accepted
CR002	ORD4044	Onigiri	2	4.75	Takeaway	Accepted
CR002	ORD2222	BBQRibs	1	24.99	Takeaway	Accepted
CR002	ORD2222	MacAndChe...	1	7.25	Takeaway	Accepted
CR002	ORD1111	JapaneseC...	1	15.25	Dine-in	Accepted

Figure 17: Order History Screen

This interface displays detailed order histories with columns for CustomerID, OrderID, ItemName, Quantity, Price, ServiceOption and Status. Users can search for orders by email using the input field at the top. The “Reorder” button allows users to easily repurchase items they previously enjoyed, streamlining the buying process and enhancing customer convenience.



Name	Email	Rating	Feedback
Anusha Devy	sha@mail.com	5	The food was ama...
Arab	arab@gmail.com	1	The food was terri...
Hello	Hello@mail.com	2	Service was terrible

Figure 18: Customer Reviews Screen

This page presents customer reviews in a tabular format, including columns for Name, Email, Rating, and Feedback. The clean, organized layout makes it easy to read feedback, which is valuable for both customers and vendors. It encourages transparency by showcasing genuine user experiences, thereby aiding other customers in making informed purchasing decisions.

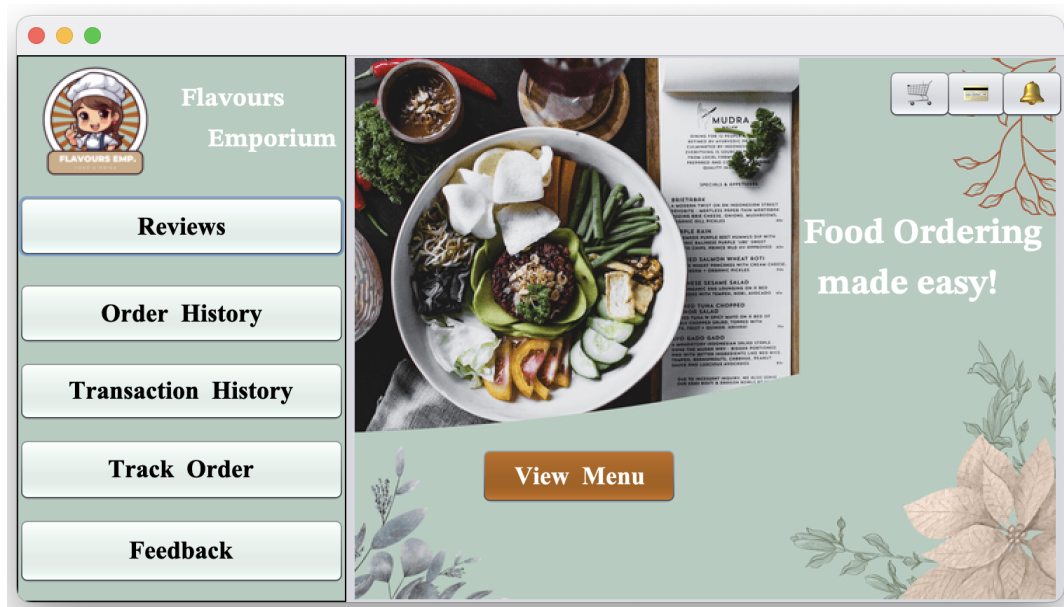


Figure 19: Home Screen

The main navigation screen for the “Flavours Emporium” app, featuring buttons for Reviews, Order History, Transaction History, Track Order, and Feedback. The visually appealing design invites users to explore the menu and other features. The shopping cart, credit balance, and notifications icons at the top right provide quick access to essential functions, creating a cohesive and user-friendly navigation experience.

2.2.3 Delivery Runner

2.2.3.1 Delivery Runner Login Page

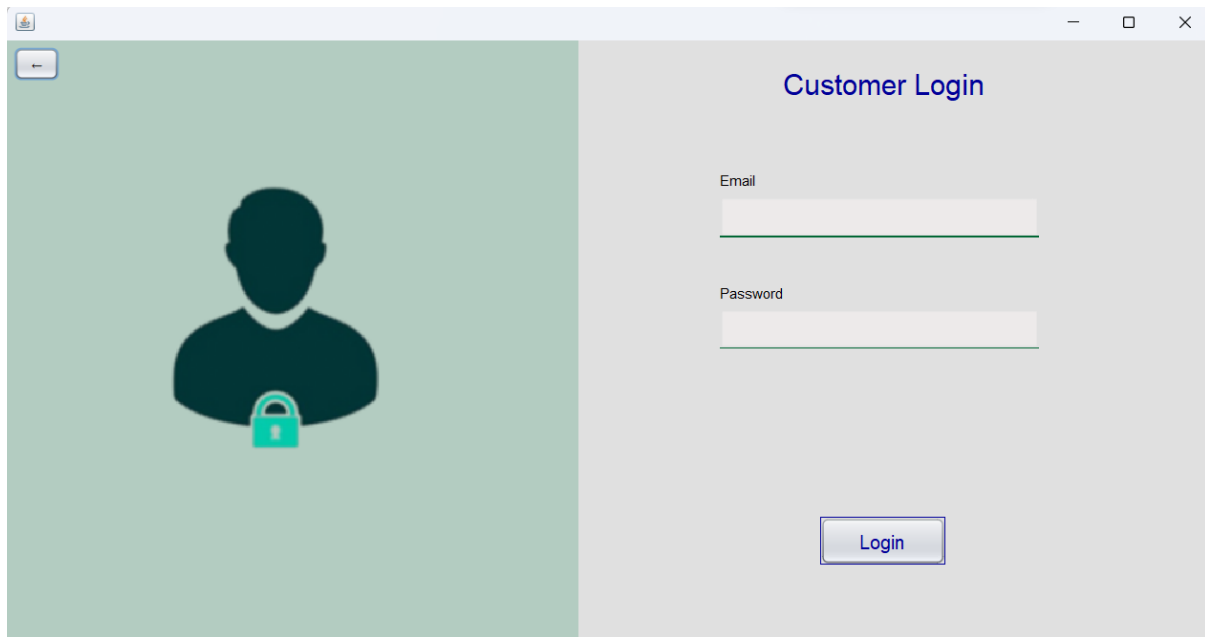


Figure 20: Delivery Runner Login Page

The Delivery Runner Login page is crafted to enable runners to securely access their accounts through the application. It features two input fields: Email and Password, which ensure that only authorized individuals can log in. The interface boasts a clean and minimalist design with the login form positioned on the right side of the screen. Upon entering credentials correctly, pressing the Login button authenticates users and grants them access to their dashboard. Additionally, there's a Back button located on the left side for easy navigation back to previous pages if needed. This robust login system safeguards sensitive information by ensuring that only verified runners have access to view tasks, update statuses, and track earnings.

2.2.3.2 Delivery Runner Dashboard

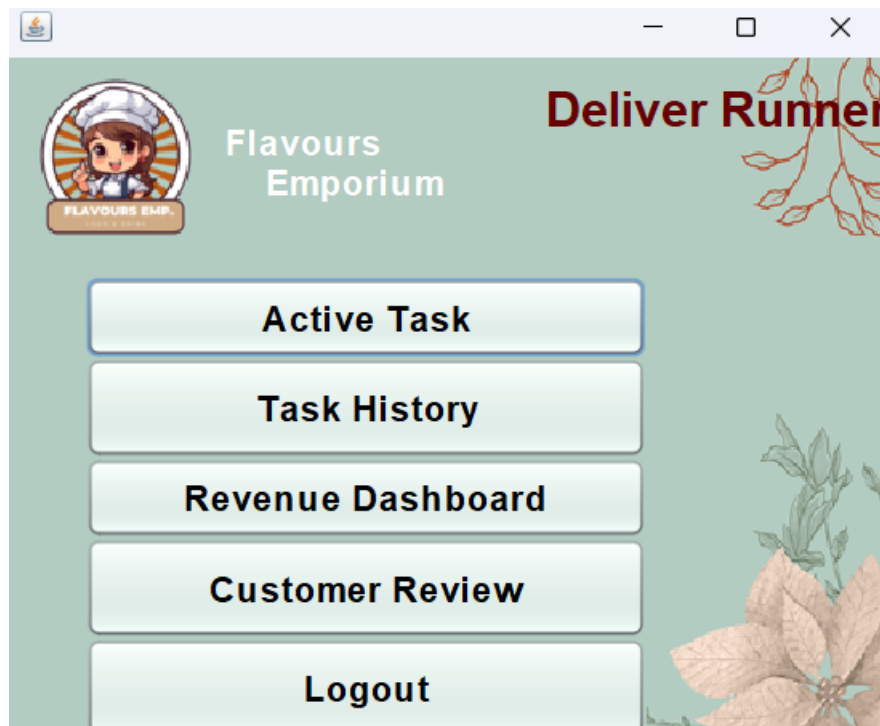
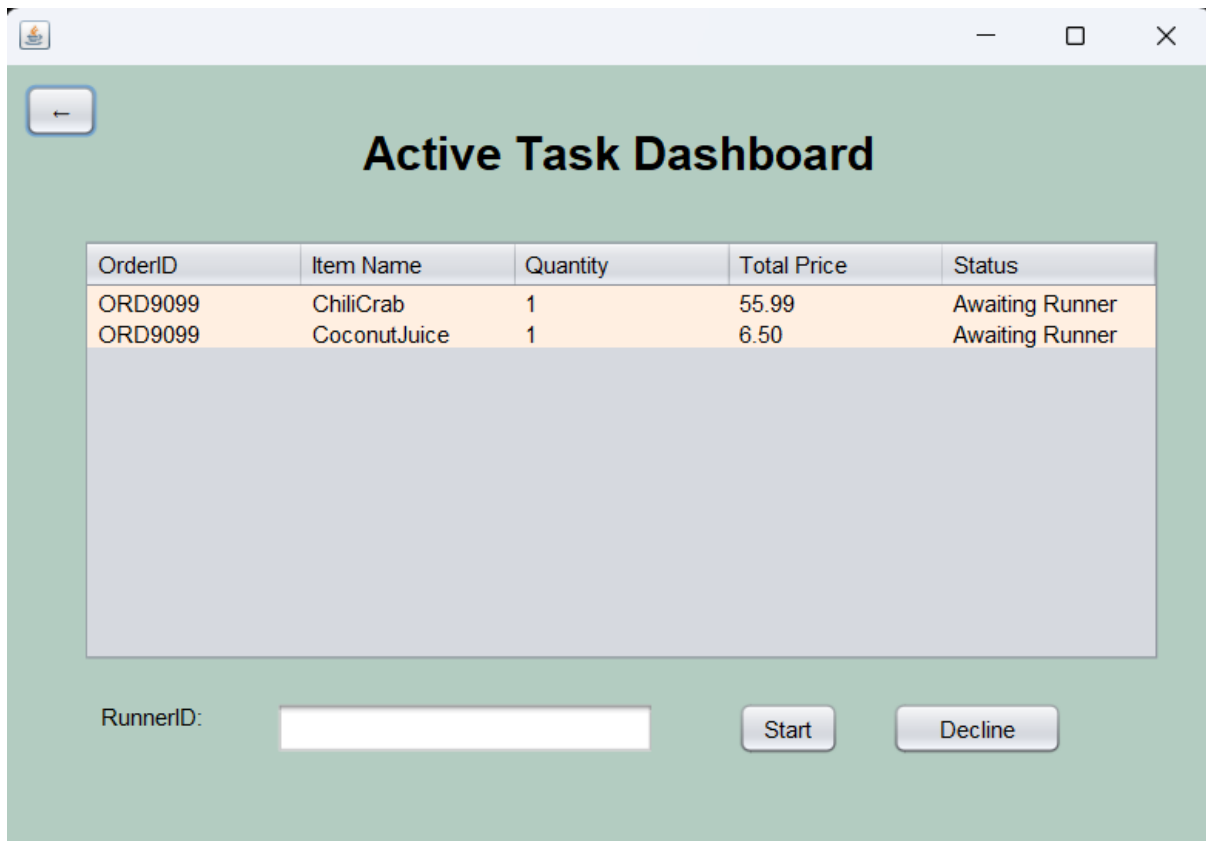


Figure 21 Delivery Runner Dashboard Output

The Delivery Runner Dashboard is a unified platform designed to simplify task management and performance monitoring for runners. It enables them to access assigned duties, choose whether to accept or decline delivery requests, and update the status of tasks instantly. The dashboard also features a task history section that allows runners to revisit completed orders. Furthermore, it includes an earnings tracker displaying income on a daily, monthly, and yearly basis so that runners can easily keep track of their financial progress. With its intuitive design and automated notification system, the dashboard boosts efficiency by ensuring smooth communication among runners, vendors, and customers.

2.2.3.3 Active Task Page



Active Task Dashboard

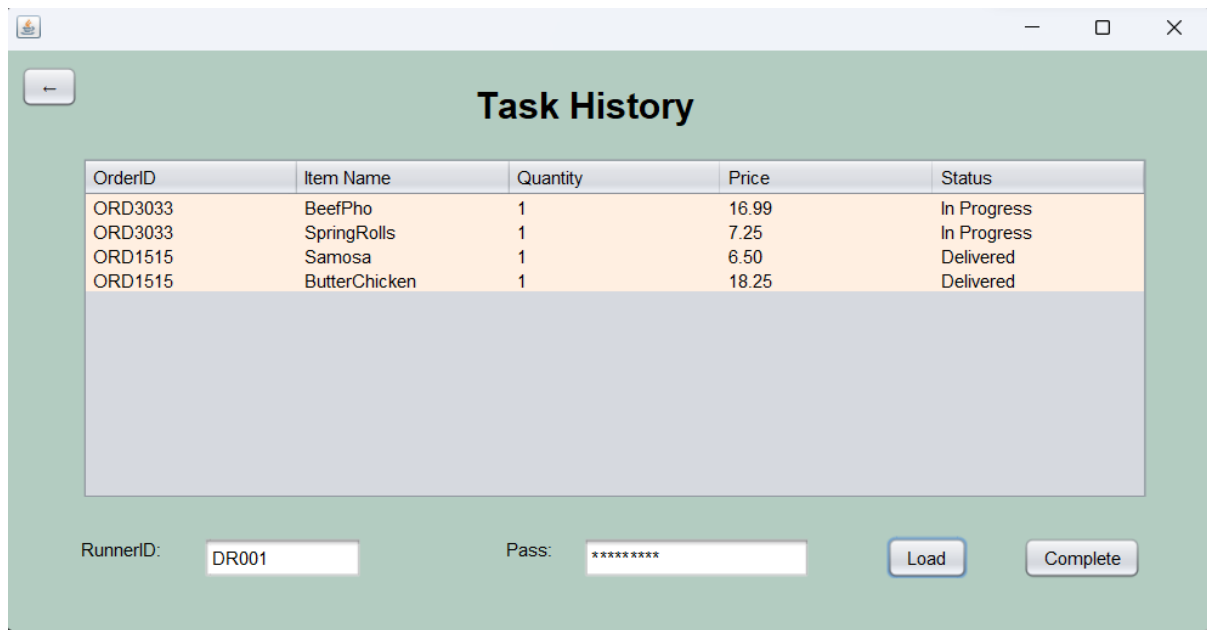
OrderID	Item Name	Quantity	Total Price	Status
ORD9099	ChiliCrab	1	55.99	Awaiting Runner
ORD9099	CoconutJuice	1	6.50	Awaiting Runner

RunnerID:

Figure 22 Active Task Page

The Active Task page serves as a dynamic interface for managing and tracking ongoing delivery tasks. It provides details such as Order ID, item names, quantities, prices, service options, and current statuses. Users can engage with the tasks by selecting "Start", "Decline," or "Back." If a runner declines a task, the system automatically tries to reassign it to another available runner; however, if no runners are available at that time customers will be notified of dine-in or takeaway alternatives. The page updates task statuses in real-time while ensuring all changes are saved in orders.txt file for accurate record-keeping purposes.

2.2.3.4 Task History Page



The screenshot shows a web application window titled "Task History". Inside the window, there is a table with the following data:

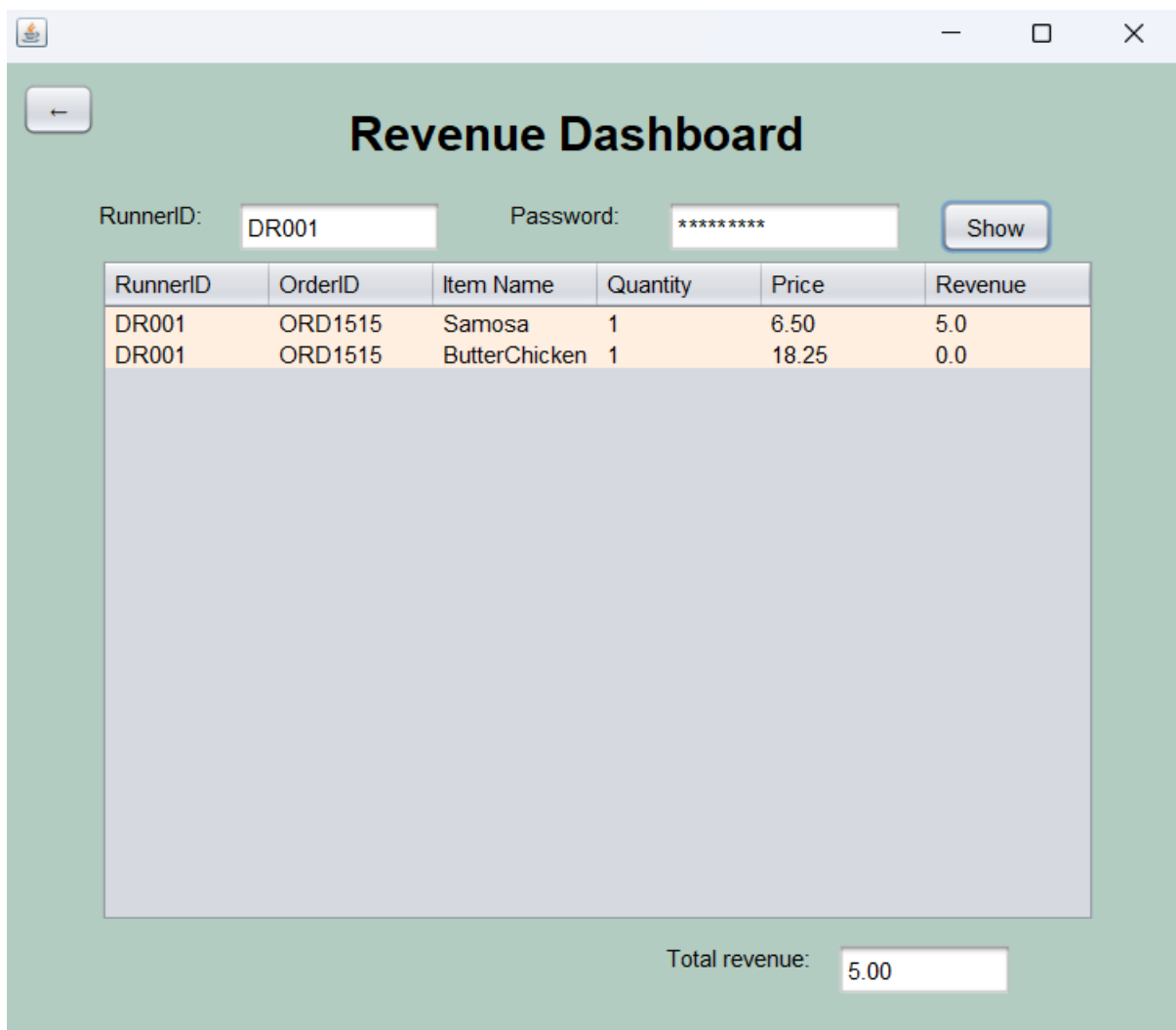
OrderID	Item Name	Quantity	Price	Status
ORD3033	BeefPho	1	16.99	In Progress
ORD3033	SpringRolls	1	7.25	In Progress
ORD1515	Samosa	1	6.50	Delivered
ORD1515	ButterChicken	1	18.25	Delivered

Below the table, there are input fields for "RunnerID:" (containing "DR001") and "Pass:" (containing "*****"). To the right of these fields are two buttons: "Load" and "Complete". A "Back" button is also visible in the top left corner of the page content area.

Figure 23 Task History Page

The Task History page showcases a delivery runner's completed tasks, offering an organized overview of previous deliveries. At the top is a table featuring columns for OrderID, Item Name, Quantity, Price, and Status; with the Status column indicating whether each task is "Delivered" or "Completed." Beneath this table are input fields for RunnerID and Password where runners can securely enter their credentials to access their task history by clicking the Load button. The Complete button is designed to update task statuses, enabling runners to mark pending items as "Completed" or "Delivered." Additionally, there is a Back button for easy navigation. This page helps runners track their work history and verify task statuses, promoting clear record-keeping and accountability.

2.2.3.4 Revenue Dashboard



Revenue Dashboard

RunnerID: Password:

RunnerID	OrderID	Item Name	Quantity	Price	Revenue
DR001	ORD1515	Samosa	1	6.50	5.0
DR001	ORD1515	ButterChicken	1	18.25	0.0

Total revenue:

Figure 24 Delivery Runner's Revenue Dashboard

The Revenue Dashboard page is crafted to showcase a delivery runner's earnings derived from completed tasks. At the top, it features fields for RunnerID and Password, enabling runners to log in securely and access their personalized revenue data. By clicking the Show button, users can fill the table below with order details including columns such as RunnerID, OrderID, Item Name, Quantity, Price, and Revenue. The Revenue column indicates earnings from each delivery while a Total Revenue field at the bottom right sums up all listed revenues. A Back button is also available for easy navigation through pages. This setup provides runners with an organized view of both individual order income and total earnings per period ensuring transparency alongside precise tracking of financial performance.

2.2.3.5 Customer Review



Name	Email	Rating	Feedback
Anusha Devy	sha@mail.com	5	The food was amazing
Arab	arab@gmail.com	1	The food was terrible

Figure 25 Customer Review Page

The Customer Reviews page showcases feedback regarding food and delivery experiences. It features a table with four columns: Name, Email, Rating, and Feedback. Customers' names and email addresses are displayed alongside their ratings which may probably on a scale from 1 to 5, as well as their written comments. The Back button located at the bottom lets users navigate to a previous page. This functionality aids businesses in collecting insights into customer satisfaction, pinpointing areas needing improvement, and ensuring high service quality.

2.2.4 Administrator

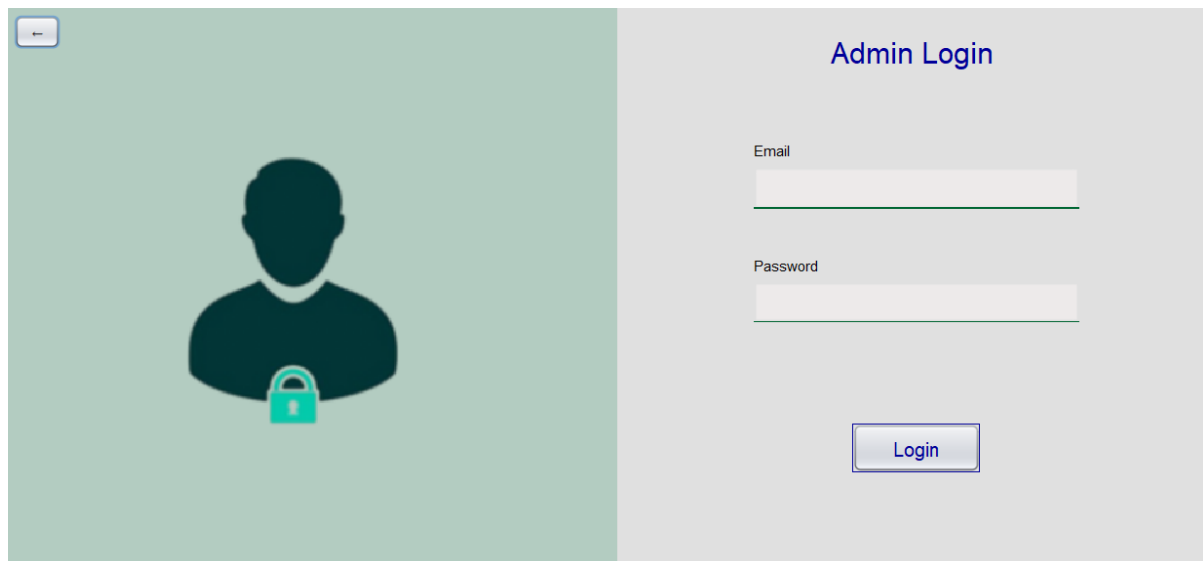
The image shows a web page for administrator login. It is split into two vertical panels. The left panel has a light green background and features a dark blue silhouette of a person's head and shoulders. Below the silhouette is a small red padlock icon with a white keyhole. The right panel has a light gray background. At the top right of this panel, the text "Admin Login" is written in blue. Below this, there are two white input fields. The first field is labeled "Email" in small black text above it. The second field is labeled "Password" in small black text above it. At the bottom of the right panel, there is a red button with the word "Login" in white text.

Figure 26: Admin Login Page

This is admin's login page. Login details are stored in the administrators.txt. If the login credentials match with the ones in the administrators.txt, then the admin will be logged into the system and brought to the Admin Home Page.

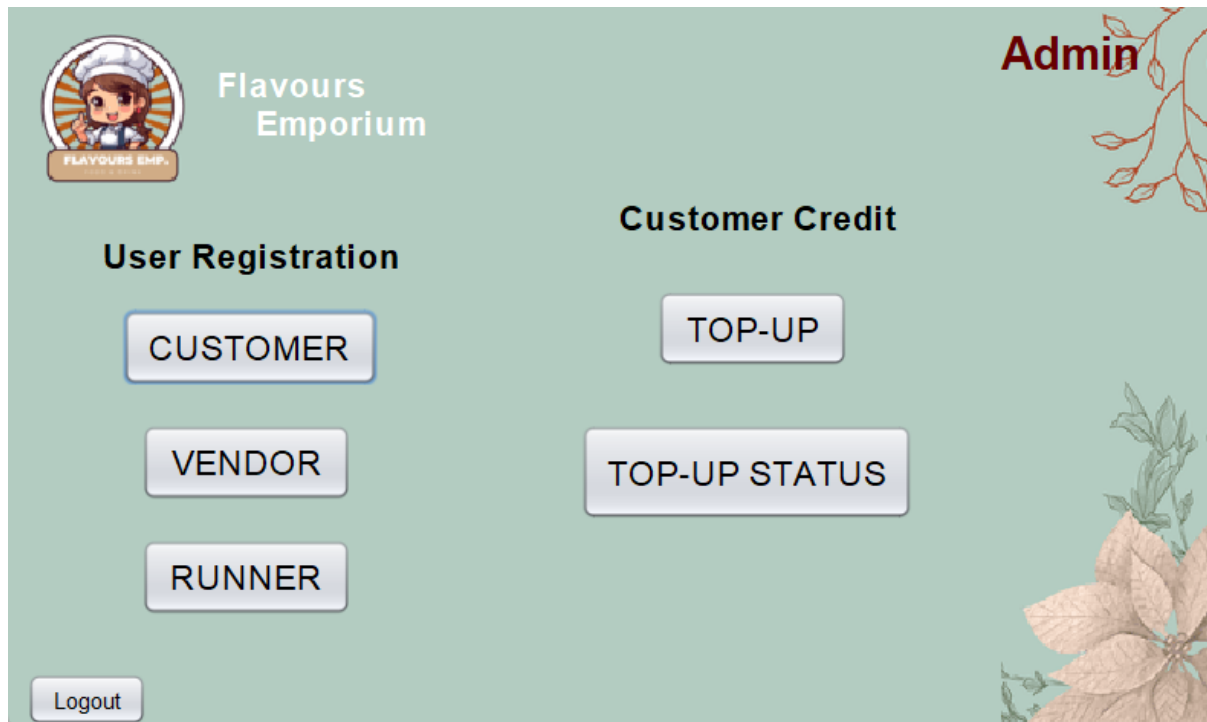


Figure 27: Admin Home Page

This is the homepage for Admin. In this page, it shows all the functionalities of the admin. Under the user registration, admin can register customers, vendors and delivery runners. For the customer credit, admin can top-up customers account and accept the request from customers who requested the admin to top up. If admin wants to logout, they can click on the logout button which will log them out of the system and bring them to the Welcome page.

CustomerID:

Name:

Email:

Password:

Contact Number:

Address:

CustomerID	Name	Email	Password	Contact Numb...	Address
CR001	Adam	adam@mail.c...	Adam@1234	0122325889	home
CR002	Hemanth	hemanth@mai...	Hemanth@12...	0186642028	abcd

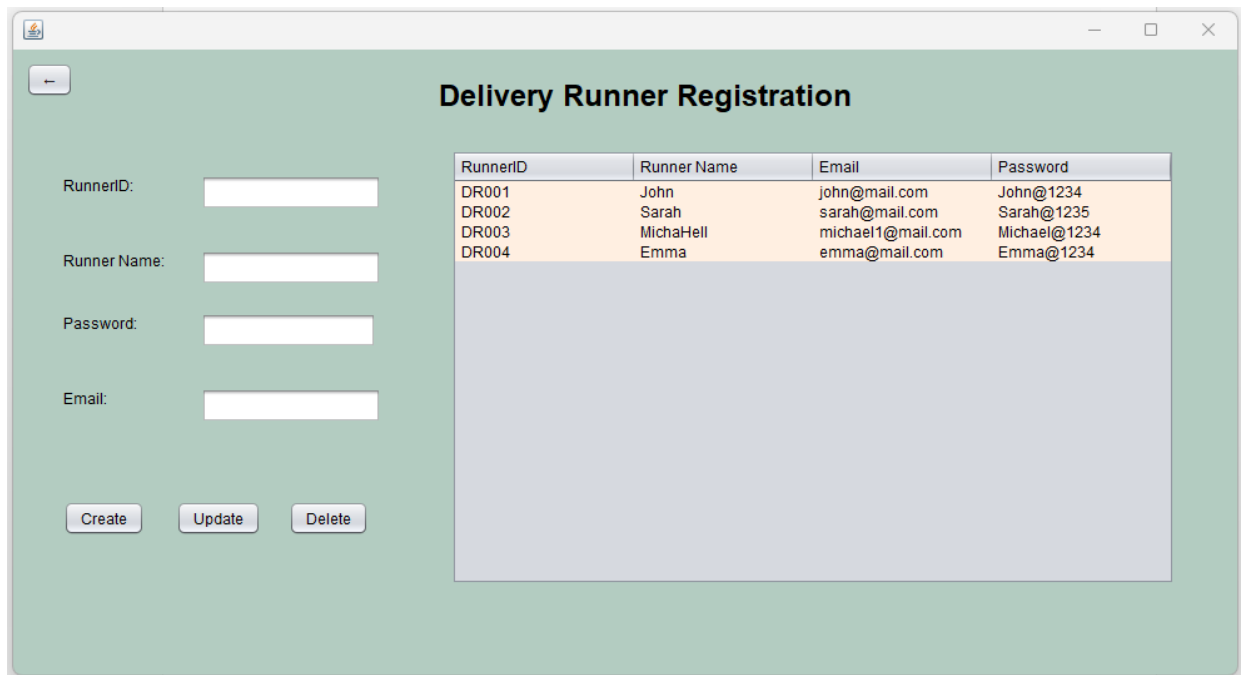
Figure 28: Customer Registration Page

This is the customer registration page. In this page, admin can register a new customer by filling in a new CustomerID for the customer together with their name, email, password, contact number and address. After creating a customer, it will automatically be saved in the customers.txt. In addition, the current balance of the new customer will automatically be RM 0.00. Besides that, admin can also update customer details from this page. Admin also can delete customers from the system.

VendorID	Email	Password	Vendor Name	Location
VR001	western@mail.com	Western@123	Western	Lot A
VR002	indian@mail.com	Western@123	Indian	Lot B

Figure 29: Vendor Registration Page

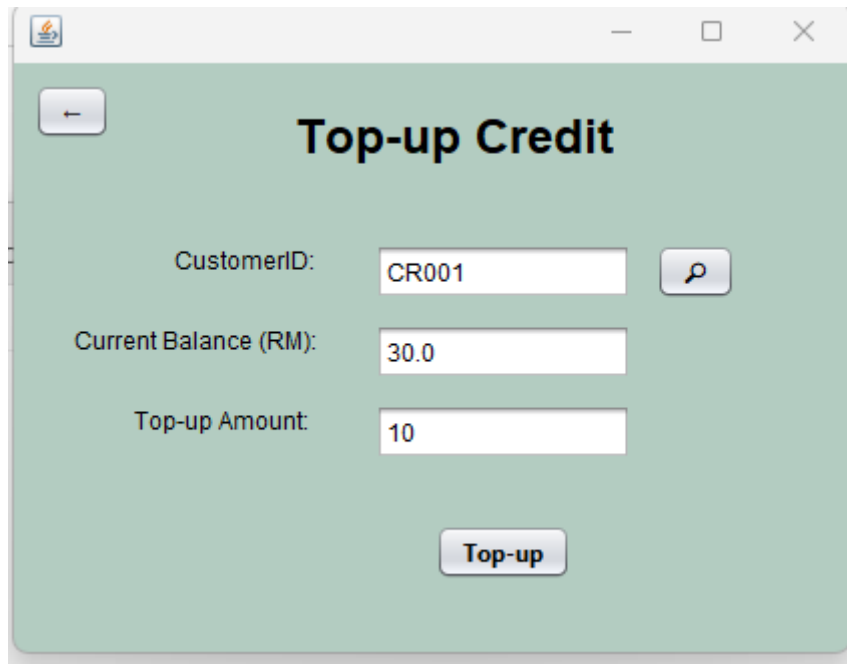
This is the vendor registration page. In this page, admin can register a new vendor by filling in a new VendorID for the vendor together with their email, password, vendor name and location. After creating a vendor, it will automatically be saved in the vendors.txt. Admin can also update the details of vendor at any given time. Admin can delete vendors from the system as well.



RunnerID	Runner Name	Email	Password
DR001	John	john@mail.com	John@1234
DR002	Sarah	sarah@mail.com	Sarah@1235
DR003	MichaHell	michael1@mail.com	Michael@1234
DR004	Emma	emma@mail.com	Emma@1234

Figure 30: Delivery Runner Registration Page

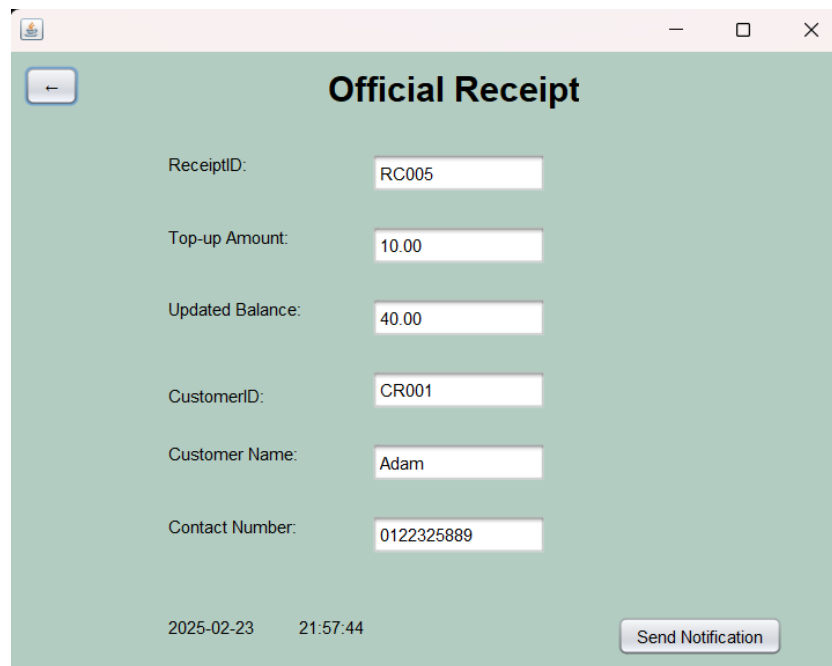
This is the delivery runner registration page. In this page, admin can register a new delivery runner in the system by providing a new RunnerID to the runner. Besides that, admin must fill up email, password and runner name. Admin can update any details of the runner in this page as well as delete any runner if necessary.



The screenshot shows a web application window titled "Top-up Credit". The interface has a light green background. At the top left, there is a small icon of a document with a flame. Below the title, there are three input fields and a search button. The first input field is labeled "CustomerID:" and contains the text "CR001". To its right is a search button with a magnifying glass icon. Below this, the second input field is labeled "Current Balance (RM):" and contains the text "30.0". Below that, the third input field is labeled "Top-up Amount:" and contains the text "10". At the bottom center, there is a button labeled "Top-up".

Figure 31: Top-up Credit page

This is the top-up credit page for the admin. In this page, admin can top-up the credit of customers. Admin must fill in the CustomerID text field, then click the search button to view the current balance of the customer. After that, admin must fill in the requested top-up amount that was requested by the customer and click on top-up. Admin will be redirected to the receipt page where it shows the breakdown of the top-up details.



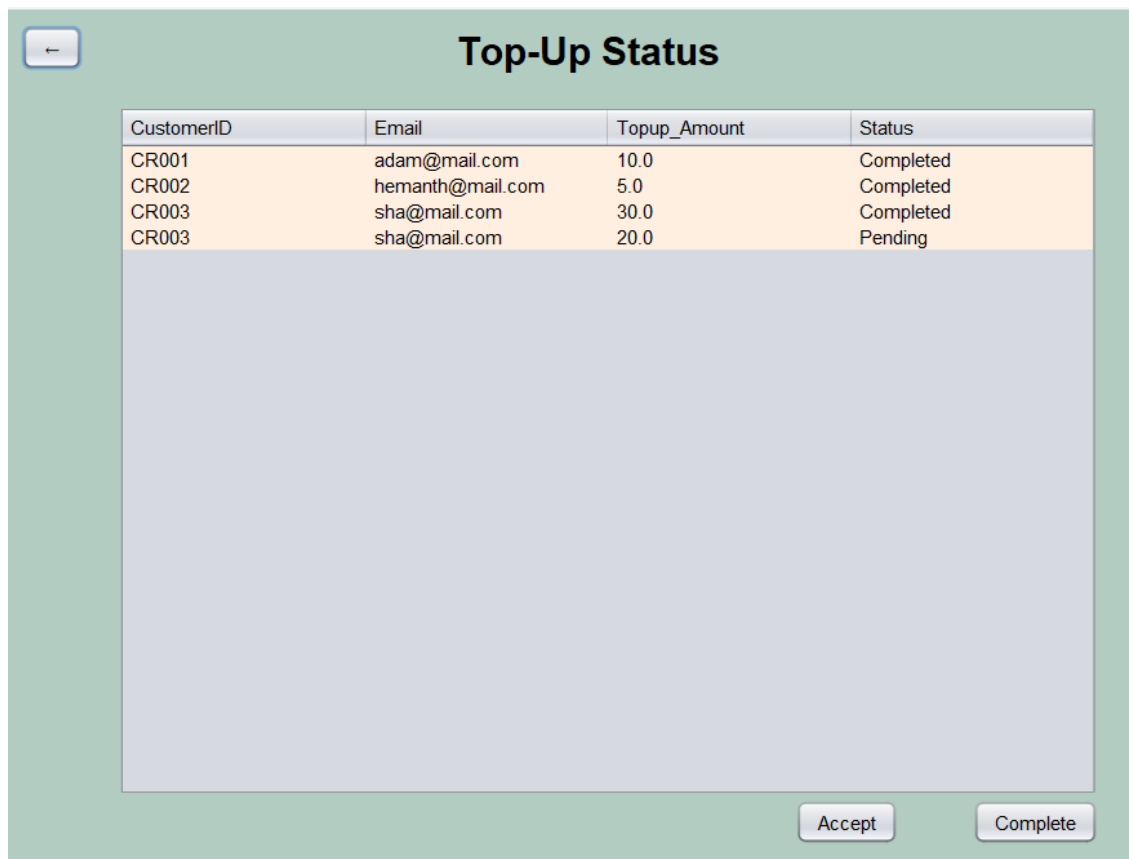
ReceiptID:	RC005
Top-up Amount:	10.00
Updated Balance:	40.00
CustomerID:	CR001
Customer Name:	Adam
Contact Number:	0122325889

2025-02-23 21:57:44

Send Notification

Figure 32: Receipt Page after Top-up

This is the official receipt created after admin has topped up for customer. In this receipt page, it shows the ReceiptID that was generated after the top-up. It also shows the top-up amount, updated balance after the top-up, the CustomerID of the customer, the customer's name and the contact number. It also shows the date and time this receipt was generated. There is also a send notification button, when you click on it, it sends a notification to the customer to notify them on the top-up that was successfully made.



CustomerID	Email	Topup_Amount	Status
CR001	adam@mail.com	10.0	Completed
CR002	hemanth@mail.com	5.0	Completed
CR003	sha@mail.com	30.0	Completed
CR003	sha@mail.com	20.0	Pending

Figure 33: Top-up Status page

This is the top-up status page for admin. In this page, admin will be able to view request sent from customers that wants to top-up their credit. Admin will then accept the request from customer and then proceed to top-up the requested amount for the customer. After that, admin will change the status from accepted to completed by clicking on the complete button.

2.2.5 Manager

2.2.5.1 Manager Login Page

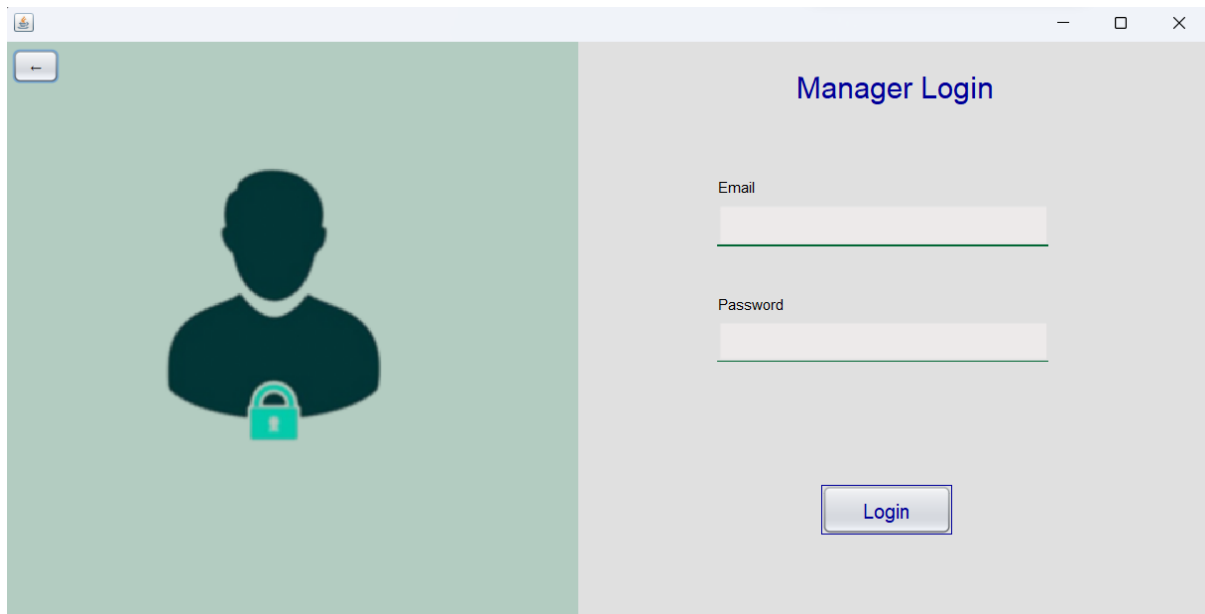


Figure 34 Manager Login Page

This Manager Login page is created to verify managers by requesting their email and password. It features fields for entering these credentials along with a "Login" button that checks the information against stored manager data. Successful credential matching grants managers access to key functionalities such as revenue tracking, task history viewing, and customer feedback management. Additionally, a back button allows users to return to the previous screen. This system safeguards sensitive business operations by ensuring only authorized individuals gain access.

2.2.5.2 Manager Home

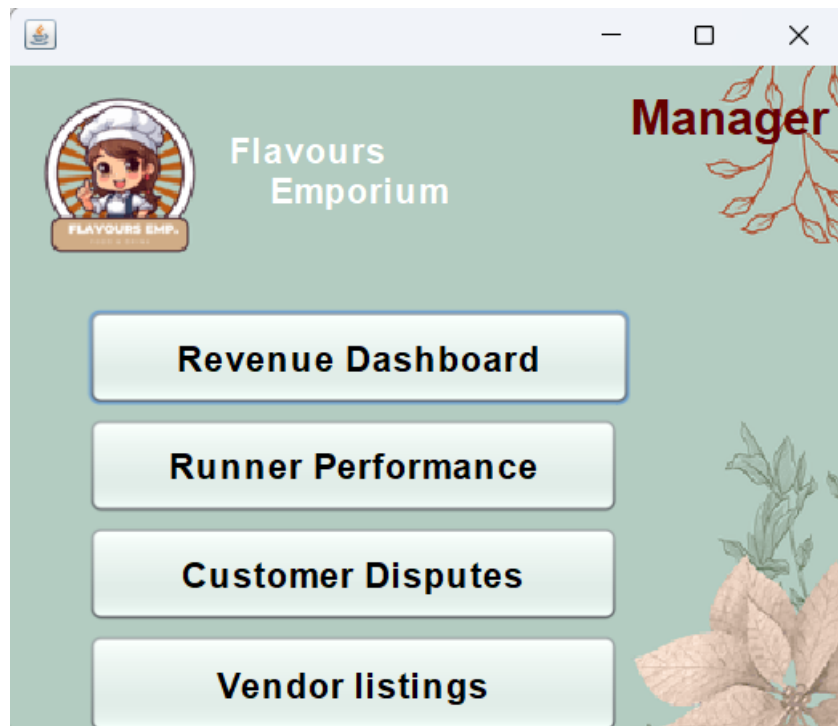
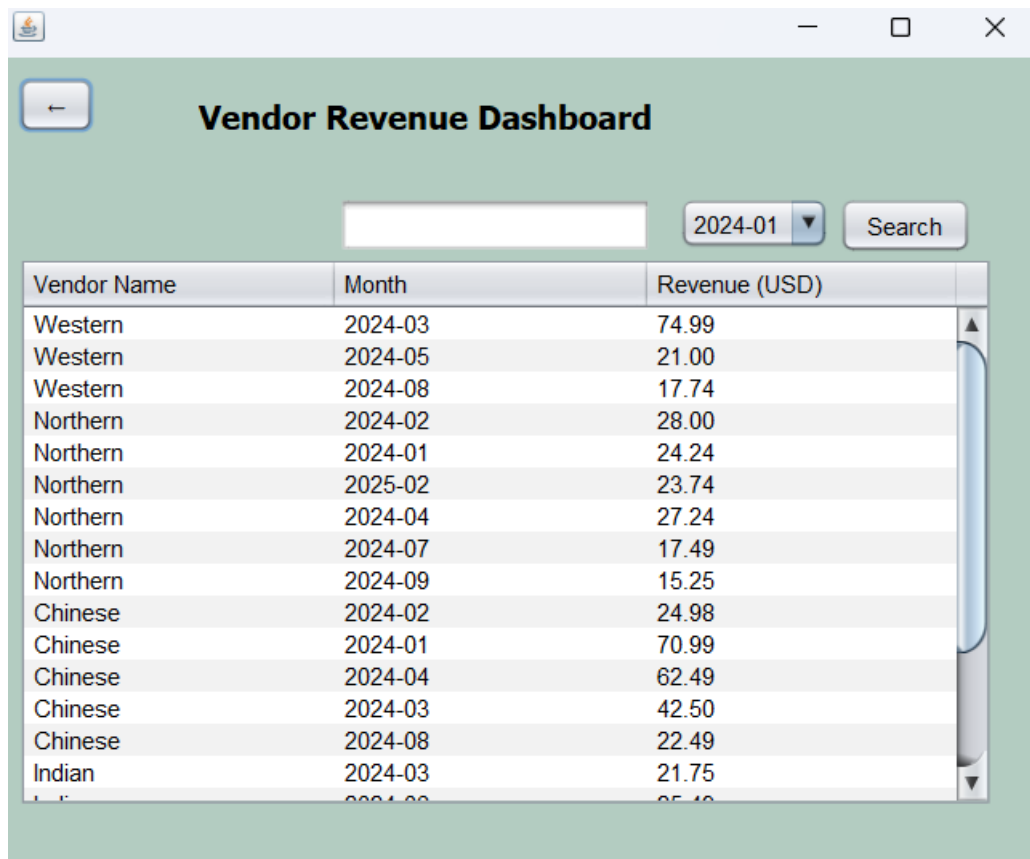


Figure 35 Manager Home Page

This Manager Home Page acts as the central hub for managers, offering easy navigation through various management tasks. It includes four primary buttons: Revenue Dashboard to monitor vendor earnings, Runner Performance to track delivery personnel efficiency, Customer Disputes for handling and resolving complaints from customers, and Vendor Listings where they can review and update vendor information. The page is visually adorned with the "Flavours Emporium" logo and boasts a clean, straightforward design. This user-friendly layout enables managers to efficiently access essential functionalities.

2.2.5.3 Vendor Revenue Dashboard



Vendor Revenue Dashboard

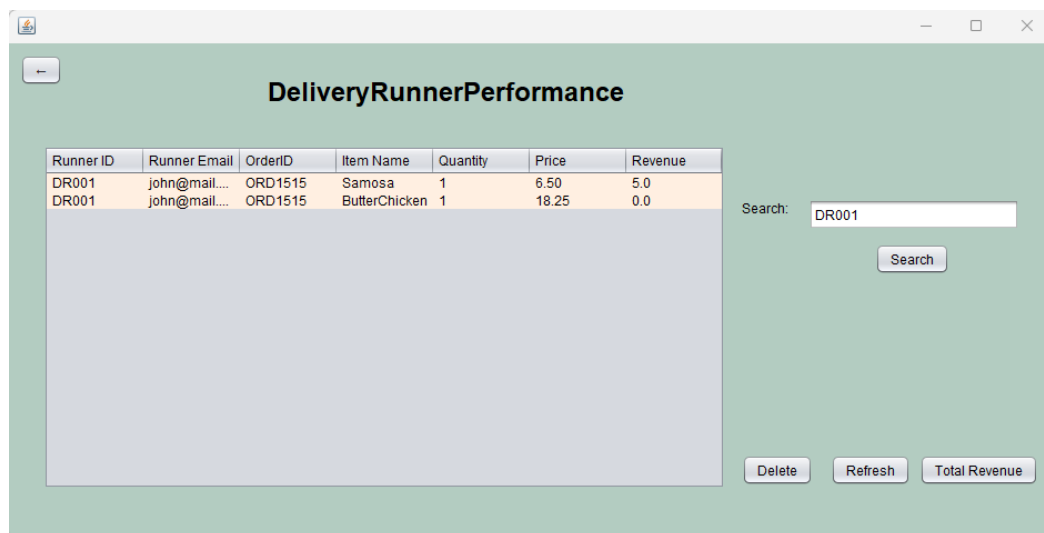
Search bar: [] Month dropdown: 2024-01 [v] Search button: Search

Vendor Name	Month	Revenue (USD)
Western	2024-03	74.99
Western	2024-05	21.00
Western	2024-08	17.74
Northern	2024-02	28.00
Northern	2024-01	24.24
Northern	2025-02	23.74
Northern	2024-04	27.24
Northern	2024-07	17.49
Northern	2024-09	15.25
Chinese	2024-02	24.98
Chinese	2024-01	70.99
Chinese	2024-04	62.49
Chinese	2024-03	42.50
Chinese	2024-08	22.49
Indian	2024-03	21.75
Indian	2024-08	25.48

Figure 36 Manager's Vendor Revenue Dashboard

This Vendor Revenue Dashboard page enables managers to monitor revenue generated by different vendors. It features a table with columns for Vendor Name, Month, and Revenue (USD), providing an easy-to-read overview of earnings over time. Managers can use a search bar to filter data by vendor names and select specific months from a dropdown menu. The "Search" button updates the results based on these filters. A back button is also available for returning to the previous screen, making it easier for managers to effectively track vendor performance.

2.2.5.4 Runner Performance Page



Runner ID	Runner Email	OrderID	Item Name	Quantity	Price	Revenue
DR001	john@mail....	ORD1515	Samosa	1	6.50	5.0
DR001	john@mail....	ORD1515	ButterChicken	1	18.25	0.0

Search:

Figure 37 Manager's Delivery Runner Performance Page

The Manager Runner Performance Page enables managers to monitor the performance of delivery runners effectively. It presents crucial information including Runner ID, Runner Email, Order ID, Item Name, Quantity, Price, and Revenue. Managers can locate specific records via a search bar and utilize buttons for Delete or Refresh functions as well as Total Revenue calculations. This page aids in evaluating runner efficiency and earnings while ensuring seamless delivery operations by tracking completed tasks and revenues generated.

2.2.5.5 Customer Disputes

CustomerID	Name	Email	Feedback	Rating
CR004	Arab	arab@gmail.c...	The food was ...	1

Email:

Customer:

Reply:

Figure 38 Manager's Customer Disputes Page

This Manager Runner Performance Page enables managers to monitor the performance of delivery runners effectively. It presents crucial information including Runner ID, Runner Email, Order ID, Item Name, Quantity, Price, and Revenue. Managers can locate specific records via a search bar and utilize buttons for Delete or Refresh functions as well as Total Revenue calculations. This page aids in evaluating runner efficiency and earnings while ensuring seamless delivery operations by tracking completed tasks and revenues generated.

2.2.5.6 Vendor Listings

The screenshot displays a web application window titled "Vendor Listings". On the left side, there are input fields for "Vendor ID:", "MenuItemID:", "ItemName:", "Price:", "Category:", and "Description:". Above these fields are three buttons: a magnifying glass (search), a plus sign (add), and a trash can (delete). The main area of the window contains a table with the following columns: MenuItemID, VendorID, ItemName, Description, Price, and Category. The table lists various menu items such as SweetAnd..., KungPao..., SpringRolls, BubbleTea, LycheeSo..., SoyMilk, PlumJuice, ChineseH..., Biryani, Samosa, GulabJam..., MangoLa..., Tteokbokki, SundubuJ..., Hotteok, Soju, PeachJuice, Chrysanth..., CapreseS..., SeafoodLi..., Bruschetta, MarinaraP..., and LemonSo... with their respective prices and categories (Food or Beverage).

MenuItemID	VendorID	ItemName	Description	Price	Category
C001	VD003	SweetAnd...	Tangy chi...	24.50	Food
C002	VD003	KungPao...	Spicy stir-f...	26.50	Food
C003	VD003	SpringRolls	Golden-fri...	18.00	Food
C017	VD003	BubbleTea	Sweet tea ...	9.50	Beverage
C018	VD003	LycheeSo...	Refreshin...	8.00	Beverage
C020	VD003	SoyMilk	Smooth a...	6.00	Beverage
C021	VD003	PlumJuice	Sweet an...	8.00	Beverage
C022	VD003	ChineseH...	Traditiona...	7.00	Beverage
I002	VD004	Biryani	Fragrant s...	26.50	Food
I003	VD004	Samosa	Crispy pa...	10.00	Food
I010	VD004	GulabJam...	Soft milk d...	9.00	Food
I011	VD004	MangoLa...	Creamy y...	25.00	Beverage
K008	VD005	Tteokbokki	Spicy rice ...	24.50	Food
K009	VD005	SundubuJ...	Soft tofu st...	37.00	Food
K010	VD005	Hotteok	Sweet pa...	25.50	Food
K011	VD005	Soju	Traditiona...	45.00	Beverage
K017	VD005	PeachJuice	Fruity and ...	21.50	Beverage
K018	VD005	Chrysanth...	Herbal tea...	20.00	Beverage
L007	VD006	CapreseS...	Fresh tom...	21.00	Food
L008	VD006	SeafoodLi...	Pasta with...	55.50	Food
L009	VD006	Bruschetta	Toasted b...	35.00	Food
L010	VD006	MarinaraP...	Tomato m...	52.00	Food
L011	VD006	LemonSo...	Fizzy soda...	16.00	Beverage
VD010	VD005	Letta	Smoothie	5.00	Beverage

Figure 39 Manager's Vendor Listing Page

This Manager's Vendor Listings Page offers an intuitive interface for overseeing items provided by vendors. It features a table that showcases crucial information, including MenuItemID, VendorID, ItemName, Description, Price, and Category. This allows managers to effortlessly view and monitor items. On the left side of the page are fields where managers can input details like Vendor ID, MenuItemID, ItemName, Price category, and Description. These make it easy to search for specific entries or update item information as needed. At the top of this page are several key action buttons: a search button helps locate particular items quickly; a plus icon facilitates adding new listings, while a trash icon enables deleting unnecessary ones. Overall tailored towards helping manage vendor menus efficiently adjusting prices when required and organizing products properly given their categories, its organized design coupled with practical tools significantly enhances workflows involved in maintaining up-to-date vendor records.

2.3 Description and Justification of Object-Oriented Concepts

2.3.1 Encapsulation

```
public class CartItem {  
    private String name;  
    private double price;  
    private int quantity;  
  
    public CartItem(String name, double price, int quantity) {  
        this.name = name;  
        this.price = price;  
        this.quantity = quantity;  
    }  
  
    public String getName() {  
        return name;  
    }  
  
    public double getPrice() {  
        return price;  
    }  
  
    public int getQuantity() {  
        return quantity;  
    }  
  
    public void setName(String name) {  
        this.name = name;  
    }  
  
    public void setPrice(double price) {  
        this.price = price;  
    }  
  
    public void setQuantity(int quantity) {  
        this.quantity = quantity;  
    }  
  
    @Override  
    public String toString() {  
        return name + "," + price + "," + quantity;  
    }  
}
```

Figure 40: CartItem Class

I have implemented encapsulation in this block of codes. I chose this because it provides a clear way to protect the internal details of the CartItem class. In my design, the variables for name, price, and quantity are marked as private, which prevents direct access from outside the class. This means any interaction with these variables must happen through the public getter and setter methods, ensuring that changes to the data are controlled and predictable. I also made sure that every CartItem is created with valid initial values by using a constructor that requires a name, price, and quantity. By doing so, each item is guaranteed to have all the necessary information as soon as it is created, reducing the chance of errors later. Finally, by overriding the toString method, I provide a straightforward way to display the item details when needed. This approach not only keeps the code tidy but also makes it easier to manage and maintain, which is why encapsulation was my chosen design strategy for this solution.

```

public class MenuItem {
    private String menuItemID;
    private String vendorID;
    private String itemName;
    private String description;
    private double price;
    private String category;

    public MenuItem(String menuItemID, String vendorID, String itemName, String description, double price, String category) {
        this.menuItemID = menuItemID;
        this.vendorID = vendorID;
        this.itemName = itemName;
        this.description = description;
        this.price = price;
        this.category = category;
    }

    public String getMenuItemID() { return menuItemID; }
    public String getVendorID() { return vendorID; }
    public String getItemName() { return itemName; }
    public String getDescription() { return description; }
    public double getPrice() { return price; }
    public String getCategory() { return category; }

    public void setMenuItemID(String menuItemID) { this.menuItemID = menuItemID; }
    public void setVendorID(String vendorID) { this.vendorID = vendorID; }
    public void setItemName(String itemName) { this.itemName = itemName; }
    public void setDescription(String description) { this.description = description; }
    public void setPrice(double price) { this.price = price; }
    public void setCategory(String category) { this.category = category; }

    @Override
    public String toString() {
        return menuItemID + "," + vendorID + "," + itemName + "," + description + "," + price + "," + category;
    }
}

```

Figure 41 :MenuItem Class

I have implemented encapsulation in this block of codes. I chose this approach because it safeguards the data within the MenuItem class. In this design, the class holds details such as menuItemID, vendorID, itemName, description, price, and category, all declared as private. This means these variables cannot be accessed directly from outside the class, ensuring that any interaction with them occurs solely through the provided public methods. By using getter and setter methods, I control how the data is read or modified, protecting the integrity of each MenuItem object. The constructor requires all essential details when a new MenuItem is created, so each object starts off with a complete set of valid information, reducing the risk of errors from incomplete data. I overrode the toString method to generate a simple, comma-separated string representation of the object, which is useful for displaying and logging item details. This encapsulated design makes the code easier to maintain and update, as any changes to the internal design will not affect other parts of the application that use the MenuItem class. I believe this encapsulation practice enhances the clarity and reliability of the code by enforcing strict boundaries between the data and its usage.

2.3.2 Abstraction

```

public DefaultTableModel getTableModel() {
    String[] columns = {"ReviewID", "CustomerID", "VendorID", "OrderID", "RunnerID", "RunnerName", "Rating", "Comments"};
    DefaultTableModel model = new DefaultTableModel(columns, 0);
    for (Review review : reviews) {
        model.addRow(new Object[]{review.getReviewID(), review.getCustomerID(), review.getVendorID(), review.getOrderID(), review.getRunnerID(), review.getRunnerName(), review.getRating(), review.getComments()});
    }
    return model;
}

public DefaultTableModel search(String query) {
    String[] columns = {"ReviewID", "CustomerID", "VendorID", "OrderID", "RunnerID", "RunnerName", "Rating", "Comments"};
    DefaultTableModel model = new DefaultTableModel(columns, 0);
    query = query.toLowerCase(); // Make the search case-insensitive

    for (Review review : reviews) {
        if (review.getReviewID().toLowerCase().contains(query) ||
            review.getCustomerID().toLowerCase().contains(query) ||
            review.getVendorID().toLowerCase().contains(query) ||
            review.getOrderID().toLowerCase().contains(query) ||
            review.getRunnerID().toLowerCase().contains(query) ||
            review.getRunnerName().toLowerCase().contains(query) ||
            String.valueOf(review.getRating()).toLowerCase().contains(query) ||
            review.getComments().toLowerCase().contains(query)) {
            model.addRow(new Object[]{
                review.getReviewID(),
                review.getCustomerID(),
                review.getVendorID(),
                review.getOrderID(),
                review.getRunnerID(),
                review.getRunnerName(),
                review.getRating(),
                review.getComments()
            });
        }
    }

    return model.getRowCount() > 0 ? model : getTableModel();
}

```

Figure 42: Abstraction OOP

The code demonstrates Abstraction by providing simplified methods, `getTableModel()` and `search()`, to manage and display review data. These methods handle complex tasks like organizing data into a table model and performing case-insensitive searches, but they hide the underlying implementation details from the user. By exposing only the necessary functionalities and keeping the internal processing logic private, the code maintains a clean and user-friendly interface. This approach makes the code easier to maintain and reuse without requiring knowledge of the inner workings.

```

25 public void loadTasks(DefaultTableModel model) {
26     model.setRowCount(0); // ✓ Clear table before loading
27
28     try (BufferedReader br = new BufferedReader(new FileReader("src/txt/DeliveryTasks.txt"))) {
29         String line;
30         while ((line = br.readLine()) != null) {
31             String[] parts = line.split(",");
32             if (parts.length >= 5) {
33                 String orderID = parts[0].trim();
34                 String itemName = parts[1].trim();
35                 String quantity = parts[2].trim();
36                 String price = parts[3].trim();
37                 String status = parts[4].trim();
38                 String assignedRunner = (parts.length > 5) ? parts[5].trim() : "";
39
40                 // ✓ Only load tasks assigned to the logged-in runner
41                 if (assignedRunner.equals(this.runnerID)) {
42                     model.addRow(new Object[]{orderID, itemName, quantity, price, status});
43                 }
44             }
45         }
46     } catch (IOException e) {
47         JOptionPane.showMessageDialog(null, "Error reading DeliveryTasks.txt file.");
48     }
49 }

```

Figure 43: Abstraction OOP (DeleiveryTask.java)

The DeliveryTask class exemplifies abstraction through methods like loadTasks(), which conceals the intricate process of file reading and task table updating. This method encapsulates the details of file handling such as opening files, reading content, and parsing data lines enabling other program components to simply call loadTasks() for results. By leveraging abstraction, this class minimizes code duplication, enhances readability, and facilitates easier maintenance; any modifications in handling files can be made within this method without impacting other sections of the program.

2.4 Additional Features

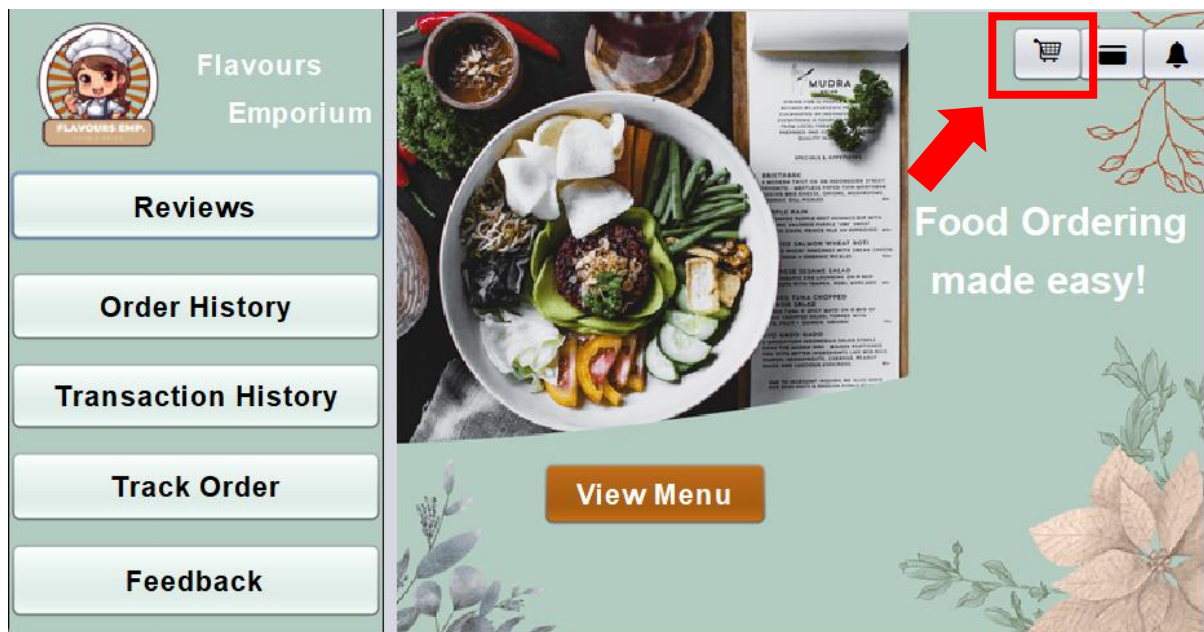


Figure 44: Customers Homepage with Cart Highlighted

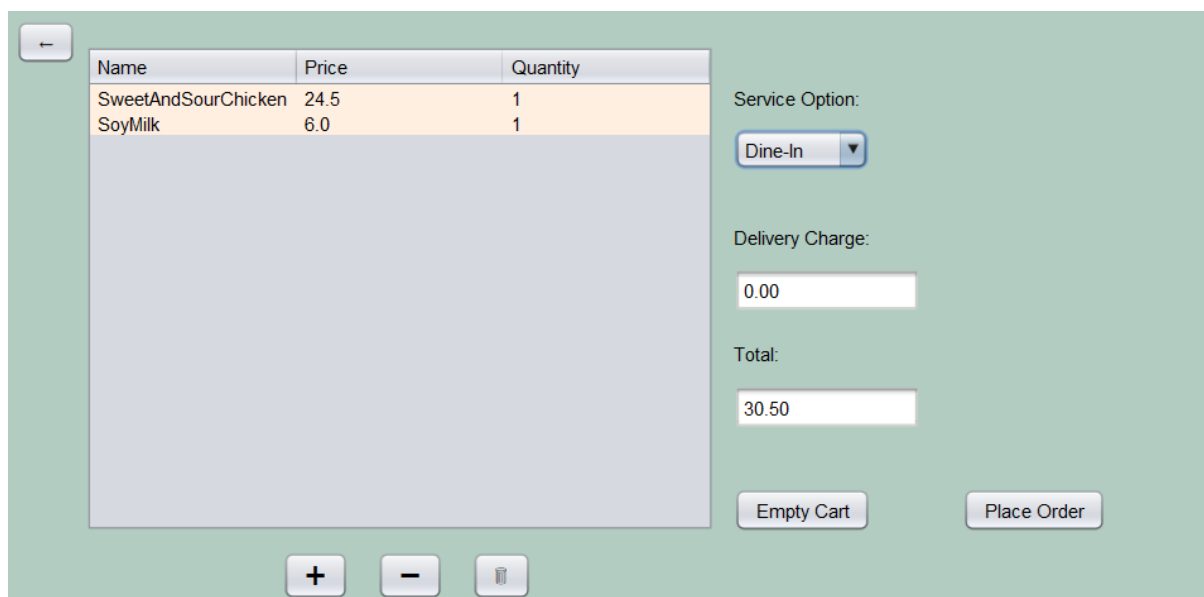


Figure 45: Customer's Cart after Ordering

Customers can add their food and drink items to the cart. This makes it easier for them to see what they have ordered and view the total amount that they have ordered. They can even select the service option and place their order from the cart itself. They can also add items from the cart area because there is an option to add or remove items from the cart page.

3.0 Limitation and Conclusion

3.1 Limitation

Although efficient in rearranging ordering of food process, Flavours Emporium has a couple of limitations. Data storage in text files limits scalability and performance, and it is difficult to manage large amounts of user and transaction data efficiently. The system is also not fully real-time capable, such as live monitoring of orders and dynamic notification, which result in notification delay and affect user experience. Security is also a concern, with no mention of encryption or other advanced security elements designed to protect sensitive data. The user interface, while functional, can be basic and lack modern design elements, which can cause lower user engagement. Furthermore, the absence of advanced analytics and reporting features hinders vendors and administrators from gaining valuable insights for business decision-making. The system also does not allow for tracking of inventory in real time, which may cause sellers to accept orders on items they do not have in hand. Another limitation is the manual credit additions made by administrators, which can be slow and particularly so during volumes. Finally, the delivery system will not cope well with effectively processing large orders, particularly where there are multiple runners out of stock or where the tasks are not redirected accordingly.

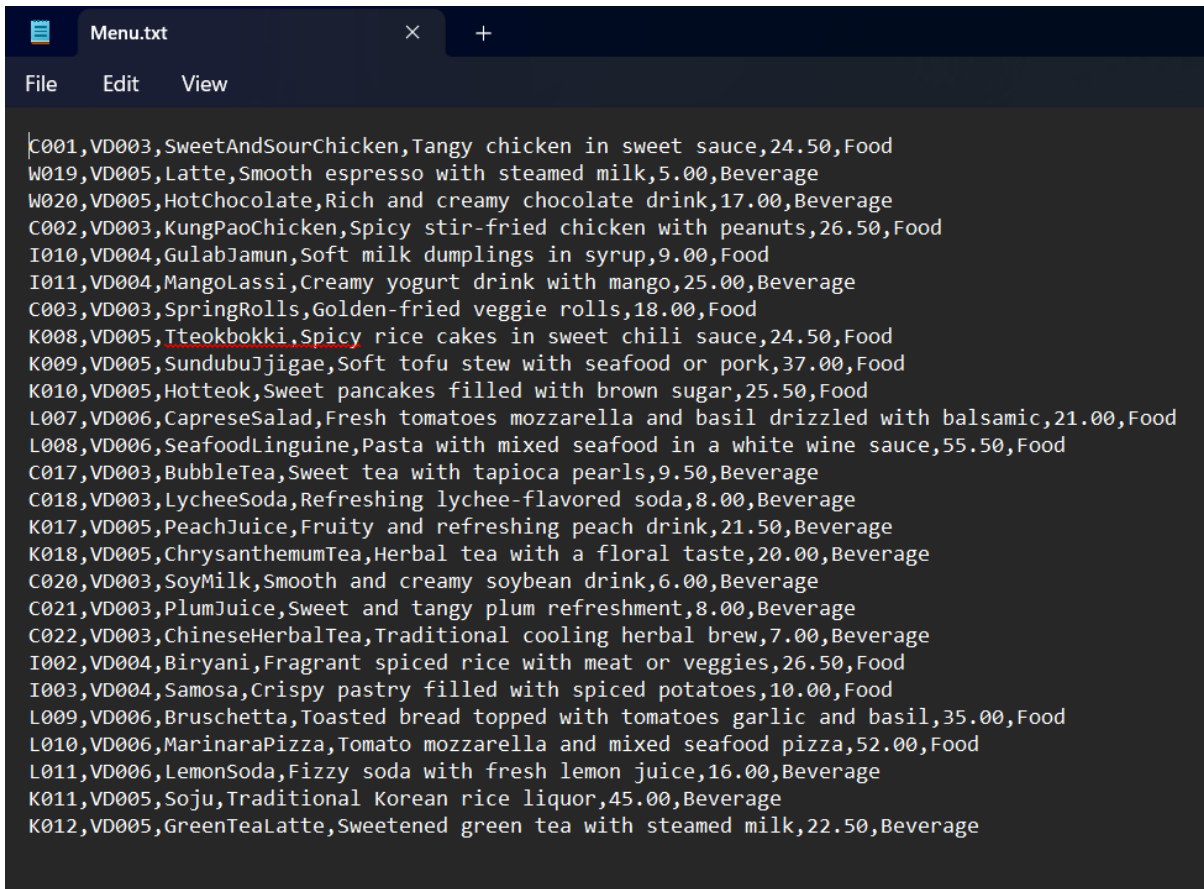
3.2 Conclusion

Flavours Emporium successfully streamlines the food ordering process by connecting customers, vendors, delivery runners, and administrators on a single platform. It allows customers to browse menus, place orders, check order status, and manage transactions with ease. Vendors can efficiently manage their menus and orders, while administrators oversee user management and credit top-ups. Developed using Java and Object-Oriented Programming (OOP) principles, the system is modular, maintainable, and easy to enhance. By using text files for data storage, it maintains simplicity in deployment while ensuring data integrity. Despite some limitations, Flavours Emporium provides a reliable and user-friendly platform that meets the basic needs of modern food ordering.

4.0 References

1. (2025). Youtube.com. <https://www.youtube.com/watch?v=uYRKNVPAC-w>
2. (2025). Youtube.com. <https://www.youtube.com/watch?v=nPFmxtRfhZk&t=1235s>
3. *Java - Character toLowerCase() Method.* (2025). Tutorialspoint.com. https://www.tutorialspoint.com/java/lang/character_tolowercase.htm
4. GeeksforGeeks. (2017, July 7). *Java program to merge two files into a third file.* GeeksforGeeks. <https://www.geeksforgeeks.org/java-program-merge-two-files-third-file/>
5. in. (2012, December 12). *I would like to create a rating system in java with GUI.* Stack Overflow. <https://stackoverflow.com/questions/13848354/i-would-like-to-create-a-rating-system-in-java-with-gui>
6. (2025). Youtube.com. https://www.youtube.com/watch?v=ZUGkQHtLS4U&list=PLyt2v1LVXYS0_-nE4ZMfJvhjDlx9kRqL

5.0 Appendix



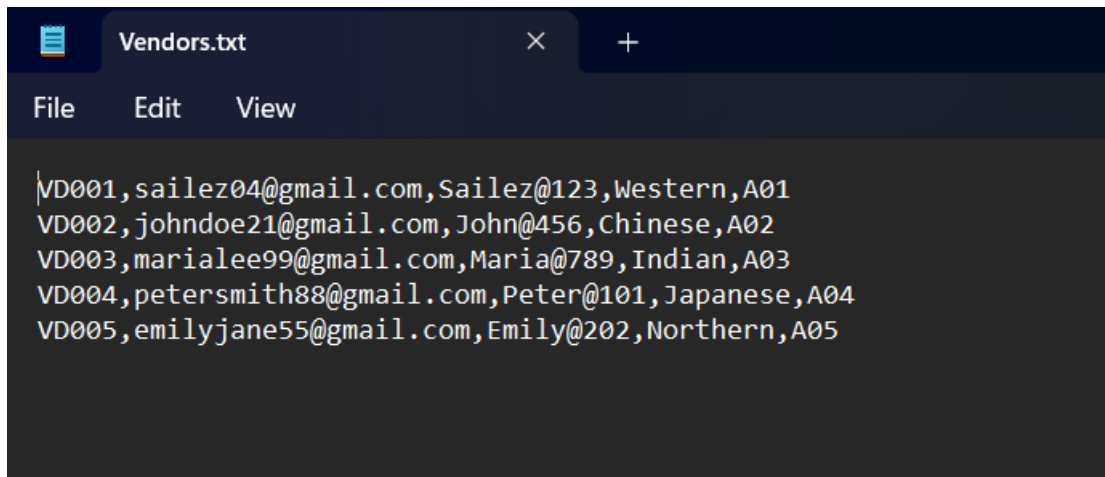
```

C001,VD003,SweetAndSourChicken,Tangy chicken in sweet sauce,24.50,Food
W019,VD005,Latte,Smooth espresso with steamed milk,5.00,Beverage
W020,VD005,HotChocolate,Rich and creamy chocolate drink,17.00,Beverage
C002,VD003,KungPaoChicken,Spicy stir-fried chicken with peanuts,26.50,Food
I010,VD004,GulabJamun,Soft milk dumplings in syrup,9.00,Food
I011,VD004,MangoLassi,Creamy yogurt drink with mango,25.00,Beverage
C003,VD003,SpringRolls,Golden-fried veggie rolls,18.00,Food
K008,VD005,Tteokbokki,Spicy rice cakes in sweet chili sauce,24.50,Food
K009,VD005,SundubuJjigae,Soft tofu stew with seafood or pork,37.00,Food
K010,VD005,Hotteok,Sweet pancakes filled with brown sugar,25.50,Food
L007,VD006,CapreseSalad,Fresh tomatoes mozzarella and basil drizzled with balsamic,21.00,Food
L008,VD006,SeafoodLinguine,Pasta with mixed seafood in a white wine sauce,55.50,Food
C017,VD003,BubbleTea,Sweet tea with tapioca pearls,9.50,Beverage
C018,VD003,LycheeSoda,Refreshing lychee-flavored soda,8.00,Beverage
K017,VD005,PeachJuice,Fruity and refreshing peach drink,21.50,Beverage
K018,VD005,ChrysanthemumTea,Herbal tea with a floral taste,20.00,Beverage
C020,VD003,SoyMilk,Smooth and creamy soybean drink,6.00,Beverage
C021,VD003,PlumJuice,Sweet and tangy plum refreshment,8.00,Beverage
C022,VD003,ChineseHerbalTea,Traditional cooling herbal brew,7.00,Beverage
I002,VD004,Biryani,Fragrant spiced rice with meat or veggies,26.50,Food
I003,VD004,Samosa,Crispy pastry filled with spiced potatoes,10.00,Food
L009,VD006,Bruschetta,Toasted bread topped with tomatoes garlic and basil,35.00,Food
L010,VD006,MarinaraPizza, Tomato mozzarella and mixed seafood pizza,52.00,Food
L011,VD006,LemonSoda,Fizzy soda with fresh lemon juice,16.00,Beverage
K011,VD005,Soju,Traditional Korean rice liquor,45.00,Beverage
K012,VD005,GreenTeaLatte,Sweetened green tea with steamed milk,22.50,Beverage

```

Figure 46 Menu.txt

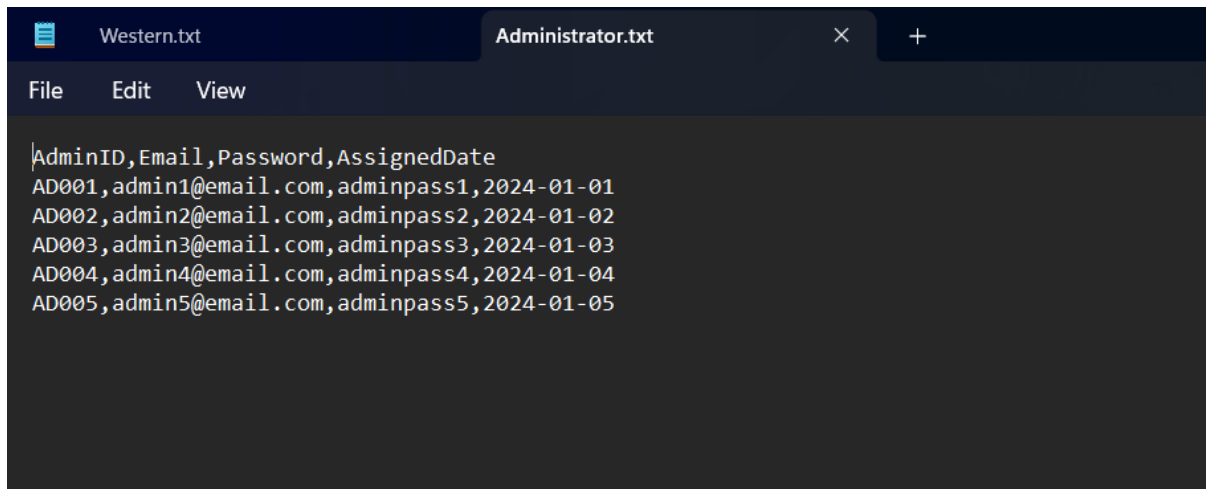
This Menu.txt file is organized with each line detailing a distinct menu item. Every record consists of several fields: MenuItemID, VendorID, ItemName, Description, Price, and Category. The Category field distinguishes items as either "Food" or "Beverage." This structured approach facilitates efficient data management and allows for smooth integration into the vendor listing system to retrieve, update, and display items easily.



```
VD001,sailez04@gmail.com,Sailez@123,Western,A01
VD002,johndoe21@gmail.com,John@456,Chinese,A02
VD003,marialee99@gmail.com,Maria@789,Indian,A03
VD004,petersmith88@gmail.com,Peter@101,Japanese,A04
VD005,emilyjane55@gmail.com,Emily@202,Northern,A05
```

Figure 47 Vendors.txt

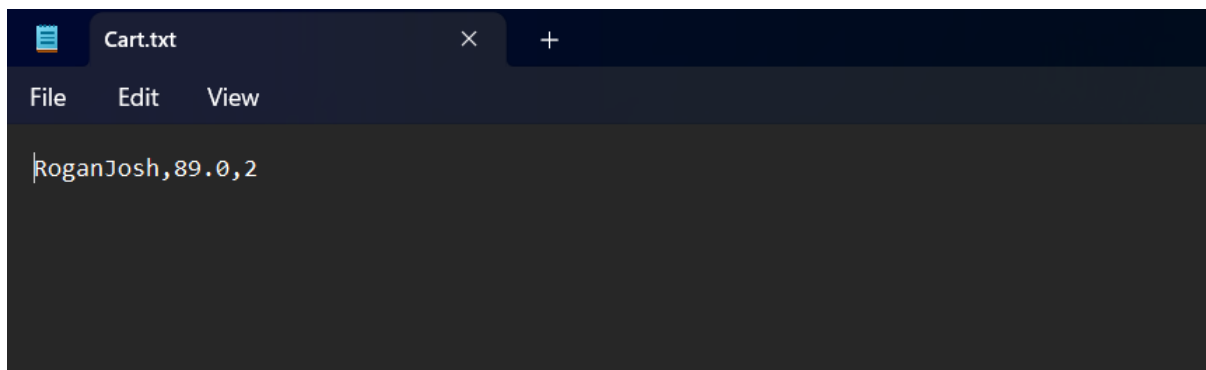
The Vendors.txt file is a structured data repository for vendors, with each line corresponding to an individual vendor. Each record comprises the following fields: VendorID, Email, Password, CuisineType, and StallNumber. This format enhances efficient management of vendors by supporting secure authentication processes, cuisine categorization capabilities, and stall assignments—thus ensuring smooth integration into the vendor listing system.



```
AdminID,Email,Password,AssignedDate
AD001,admin1@email.com,adminpass1,2024-01-01
AD002,admin2@email.com,adminpass2,2024-01-02
AD003,admin3@email.com,adminpass3,2024-01-03
AD004,admin4@email.com,adminpass4,2024-01-04
AD005,admin5@email.com,adminpass5,2024-01-05
```

Figure 48 Administrator.txt

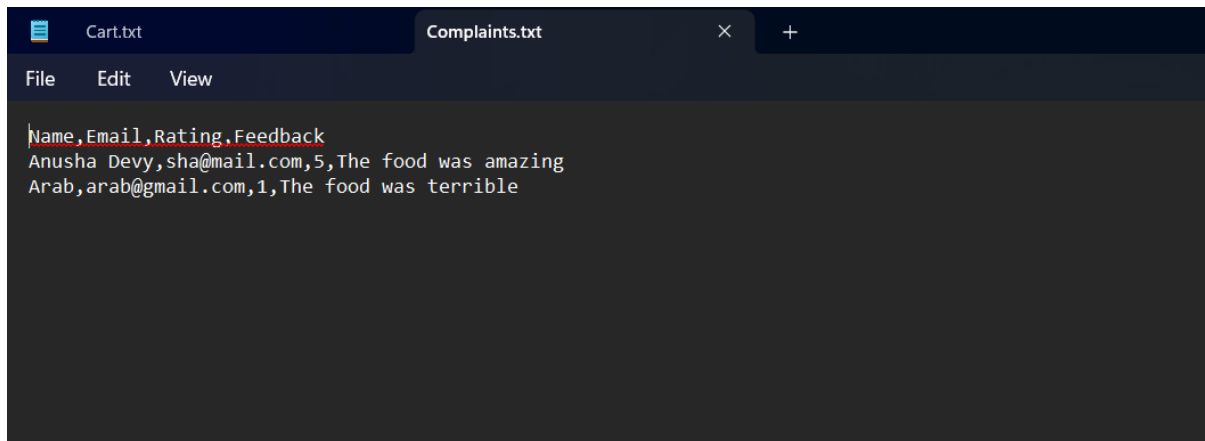
The Administrator.txt file stores organized data for administrators, with every entry comprising these fields: AdminID, Email, Password, and AssignedDate. This format aids efficient user management by enabling administrator authentication and monitoring assignments while keeping records of each admin's assignment date, ensuring well-structured and secure administrative operations.



```
RoganJosh,89.0,2
```

Figure 49 cart.txt

The Cart.txt file holds details of items a user has added to their cart, listing each entry with the ItemName, Price, and Quantity. This format facilitates straightforward cart management by supporting price calculations and quantity adjustments while ensuring accurate recording of selected items for checkout processing.



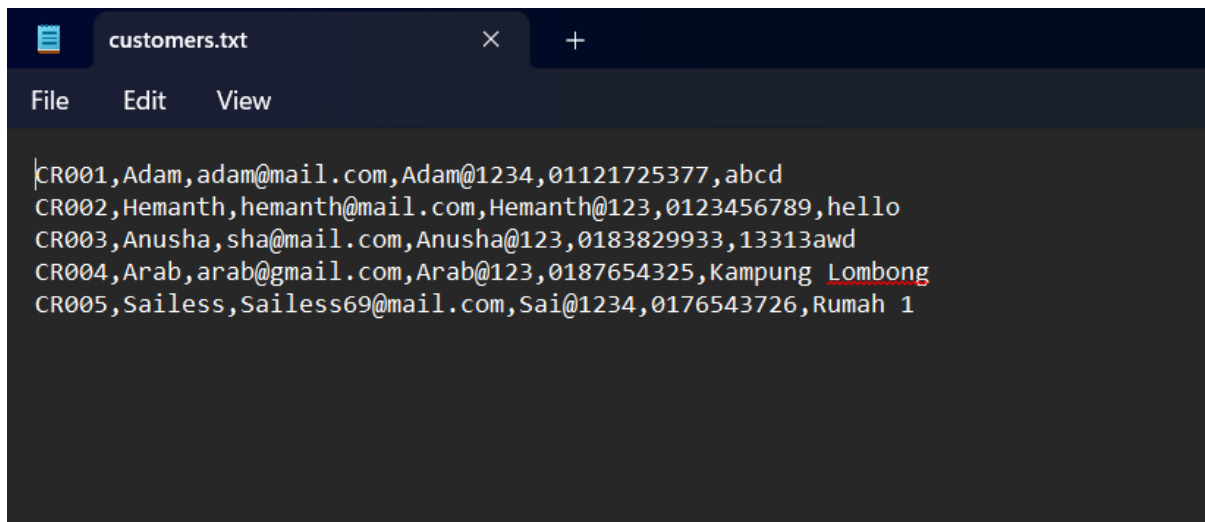
```

Name,Email,Rating,Feedback
Anusha Devy,sha@mail.com,5,The food was amazing
Arab,arab@gmail.com,1,The food was terrible

```

Figure 50 Complaiants.txt

The Complaints.txt file houses customer feedback data, with each entry consisting of the Name, Email, Rating, and Feedback. This structure aids in gathering and organizing user reviews, enabling businesses to evaluate customer satisfaction, address concerns effectively, and enhance services by relying on direct input from their clientele.



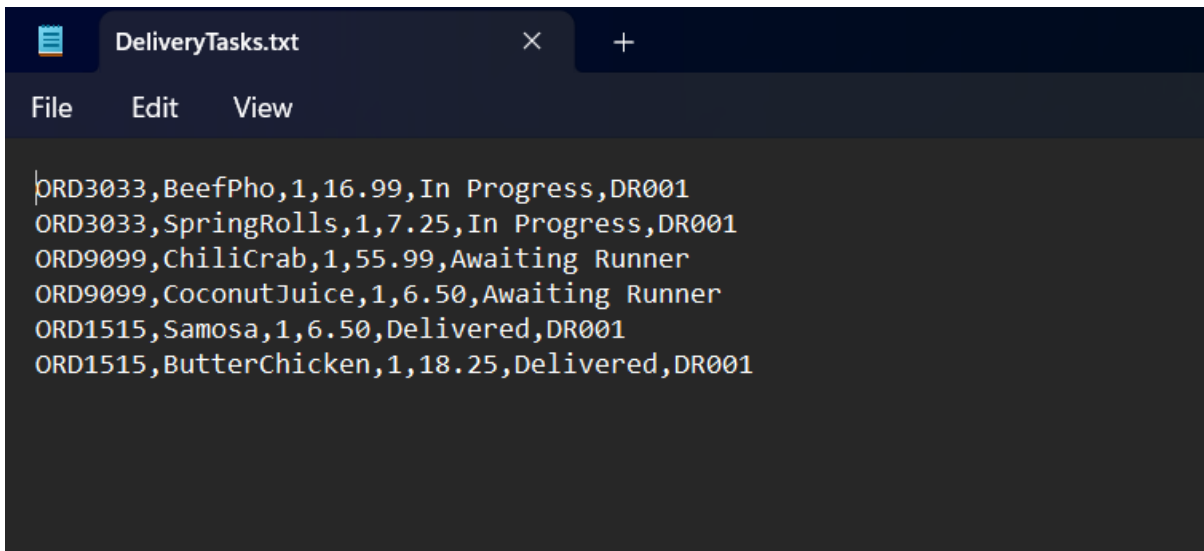
```

CR001,Adam,adam@mail.com,Adam@1234,01121725377,abcd
CR002,Hemanth,hemanth@mail.com,Hemanth@123,0123456789,hello
CR003,Anusha,sha@mail.com,Anusha@123,0183829933,13313awd
CR004,Arab,arab@gmail.com,Arab@123,0187654325,Kampung Lombong
CR005,Sailess,Sailess69@mail.com,Sai@1234,0176543726,Rumah 1

```

Figure 51 customers.txt

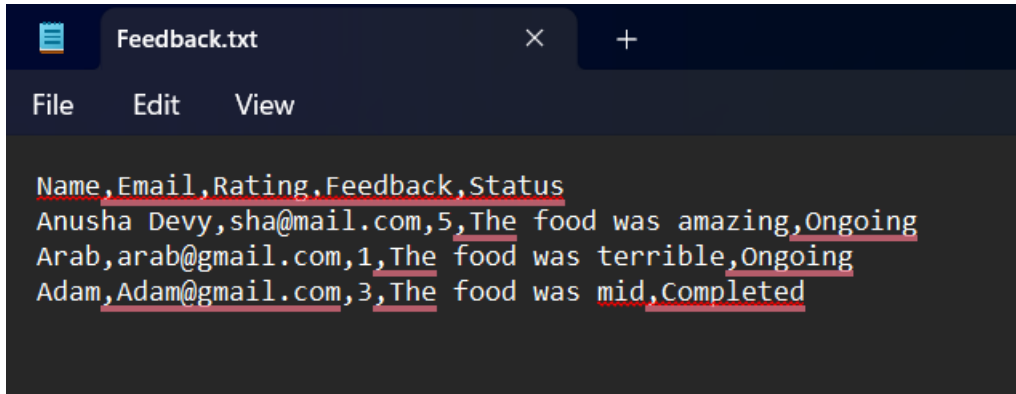
The customers.txt file holds customer details with each entry comprising CustomerID, Name, Email, Password, PhoneNumber, and Address. This structure is crucial for managing customer information as it facilitates user authentication and enables personalized contact and service. It aids business operations such as order tracking, customer support assistance, and targeted communication efforts.



```
ORD3033,BeefPho,1,16.99,In Progress,DR001
ORD3033,SpringRolls,1,7.25,In Progress,DR001
ORD9099,ChiliCrab,1,55.99,Awaiting Runner
ORD9099,CoconutJuice,1,6.50,Awaiting Runner
ORD1515,Samosa,1,6.50,Delivered,DR001
ORD1515,ButterChicken,1,18.25,Delivered,DR001
```

Figure 52 DeliveryTask.txt

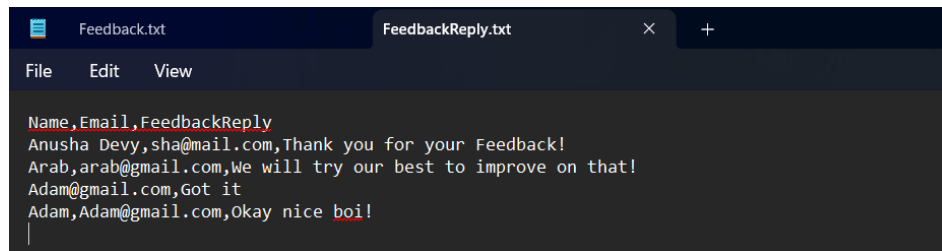
The DeliveryTasks.txt file holds details of delivery tasks, with each entry comprising OrderID, ItemName, Quantity, Price, Status, and RunnerID. This format aids in managing ongoing deliveries as well as those pending or completed by facilitating efficient task allocation and status tracking while coordinating runners to ensure seamless delivery operations.



```
Name,Email,Rating,Feedback,Status
Anusha Devy,sha@mail.com,5,The food was amazing,Ongoing
Arab,arab@gmail.com,1,The food was terrible,Ongoing
Adam,Adam@gmail.com,3,The food was mid,Completed
```

Figure 53 Feedback.txt

The Feedback.txt file holds customer feedback with the fields: Name, Email, Rating, Feedback, and Status. This setup enables the system to gather customer opinions, evaluate their experiences, and monitor whether feedback is ongoing or completed. It aids businesses in addressing customer concerns and enhancing their services.



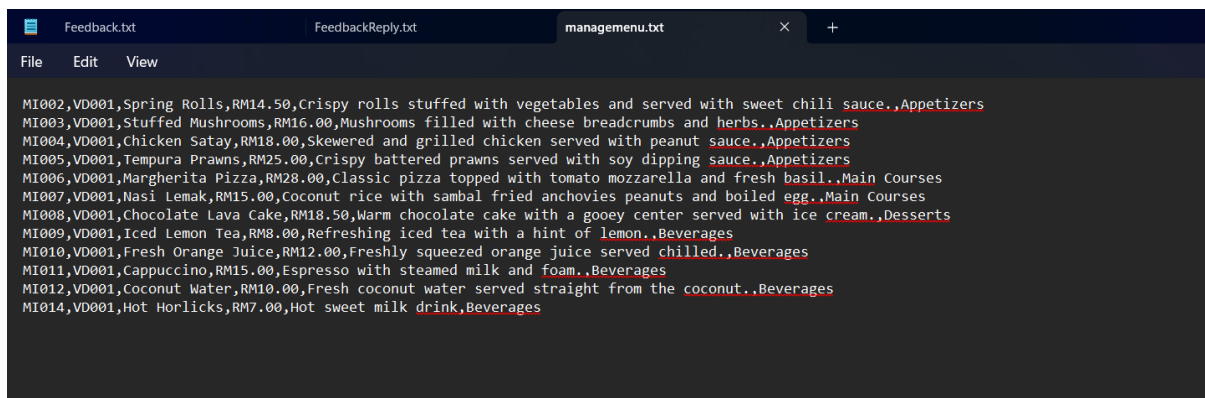
```

Name,Email,FeedbackReply
Anusha Devy,sha@mail.com,Thank you for your Feedback!
Arab,arab@gmail.com,We will try our best to improve on that!
Adam@gmail.com,Got it
Adam,Adam@gmail.com,Okay nice boil

```

Figure 54 FeedbackReply.txt

The FeedbackReply.txt file holds responses to customer feedback and includes the fields: Name, Email, and Feedback Reply. This file enables the system to document replies to customers' input, ensuring proper communication and follow-up. It provides a structured approach for addressing customer concerns, acknowledging their contributions, and enhancing overall satisfaction.



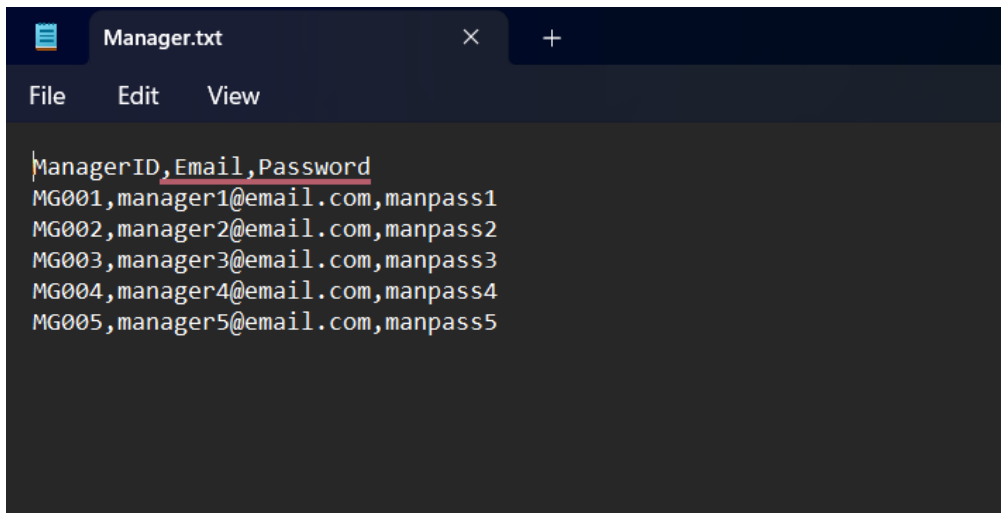
```

MI002,VD001,Spring Rolls,RM14.50,Crispy rolls stuffed with vegetables and served with sweet chili sauce.,Appetizers
MI003,VD001,Stuffed Mushrooms,RM16.00,Mushrooms filled with cheese breadcrumbs and herbs.,Appetizers
MI004,VD001,Chicken Satay,RM18.00,Skewered and grilled chicken served with peanut sauce.,Appetizers
MI005,VD001,Tempura Prawns,RM25.00,Crispy battered prawns served with soy dipping sauce.,Appetizers
MI006,VD001,Margherita Pizza,RM28.00,Classic pizza topped with tomato mozzarella and fresh basil.,Main Courses
MI007,VD001,Nasi Lemak,RM15.00,Coconut rice with sambal fried anchovies peanuts and boiled egg.,Main Courses
MI008,VD001,Chocolate Lava Cake,RM18.50,Warm chocolate cake with a gooey center served with ice cream.,Desserts
MI009,VD001,Iced Lemon Tea,RM8.00,Refreshing iced tea with a hint of lemon.,Beverages
MI010,VD001,Fresh Orange Juice,RM12.00,Freshly squeezed orange juice served chilled.,Beverages
MI011,VD001,Cappuccino,RM15.00,Espresso with steamed milk and foam.,Beverages
MI012,VD001,Coconut Water,RM10.00,Fresh coconut water served straight from the coconut.,Beverages
MI014,VD001,Hot Horlicks,RM7.00,Hot sweet milk drink.,Beverages

```

Figure 55 ManageMenu.txt

The managemenu.txt file systematically organizes the restaurant's menu by detailing each item with its pertinent attributes. Each entry includes an ItemID, serving as a unique identifier for both dishes and beverages, along with a VendorID to categorize them appropriately. The ItemName clearly indicates the name of each menu offering, while the Price is denoted in Malaysian Ringgit using "RM". A concise yet enticing Description provides insight into key ingredients or preparation techniques—such as crispy rolls paired with sweet chili sauce or espresso complemented by steamed milk and foam. Additionally, items are classified under specific Categories like Appetizers, Main Courses, Desserts, or Beverages to maintain clarity and ease of navigation within the menu structure. This format streamlines adding new entries, making updates simple, enhancing display efficiency which ultimately bolsters smooth restaurant operations.

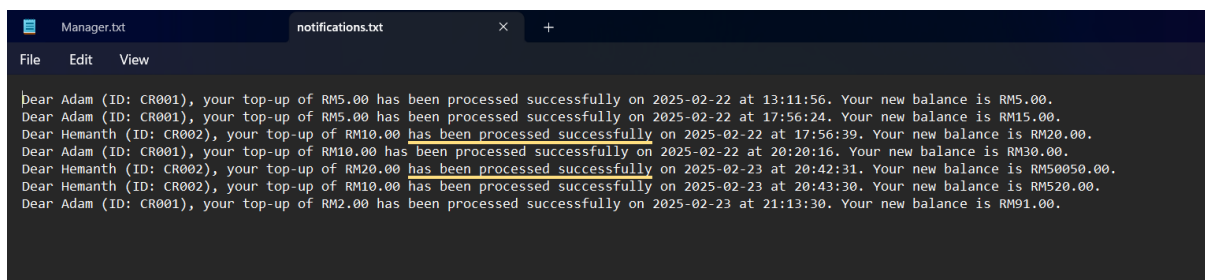


```

ManagerID,Email,Password
MG001,manager1@email.com,manpass1
MG002,manager2@email.com,manpass2
MG003,manager3@email.com,manpass3
MG004,manager4@email.com,manpass4
MG005,manager5@email.com,manpass5
    
```

Figure 56 Managers.txt

The Manager.txt file thoroughly manages manager login authorization by saving essential details in a controlled format. Each entry includes a ManagerID, which entirely finds each manager, an Email used for verification and email, and a Password that ensures secure access. The data is placed in a comma-separated format, letting on behalf of easy parsing and getting the login information when needed. This constructed method shortens the authentication process and maintains an ordered record-keeping system for managerial accounts.

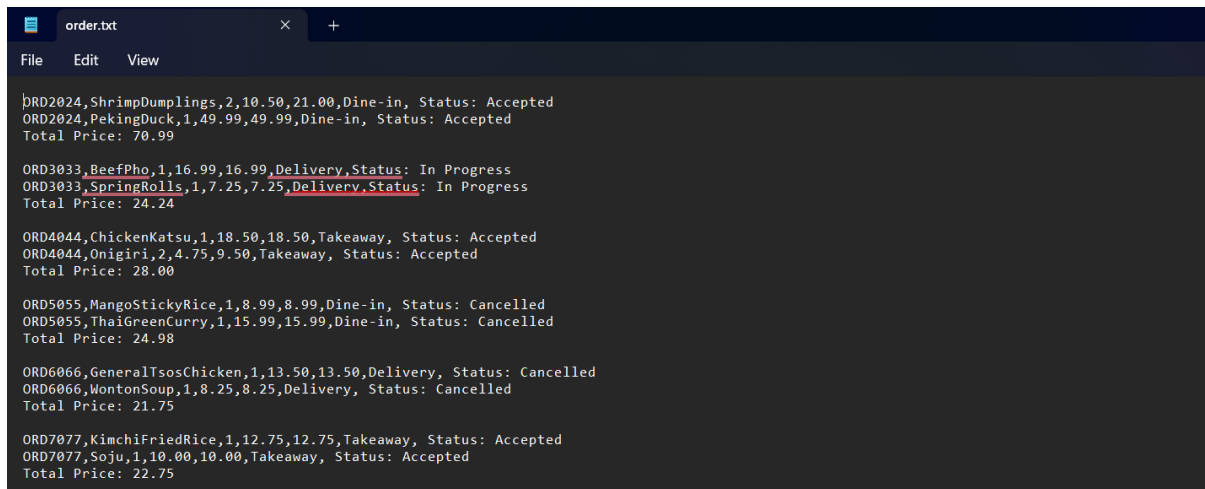


```

Dear Adam (ID: CR001), your top-up of RM5.00 has been processed successfully on 2025-02-22 at 13:11:56. Your new balance is RM5.00.
Dear Adam (ID: CR001), your top-up of RM5.00 has been processed successfully on 2025-02-22 at 17:56:24. Your new balance is RM15.00.
Dear Hemanth (ID: CR002), your top-up of RM10.00 has been processed successfully on 2025-02-22 at 17:56:39. Your new balance is RM20.00.
Dear Adam (ID: CR001), your top-up of RM10.00 has been processed successfully on 2025-02-22 at 20:20:16. Your new balance is RM30.00.
Dear Hemanth (ID: CR002), your top-up of RM20.00 has been processed successfully on 2025-02-23 at 20:42:31. Your new balance is RM50050.00.
Dear Hemanth (ID: CR002), your top-up of RM10.00 has been processed successfully on 2025-02-23 at 20:43:30. Your new balance is RM520.00.
Dear Adam (ID: CR001), your top-up of RM2.00 has been processed successfully on 2025-02-23 at 21:13:30. Your new balance is RM91.00.
    
```

Figure 57 Notifications.txt

The notifications.txt file declares a record of successful top-up transactions by customers with transparency in the balance update. Each entry consists of the name of the recipient and customer ID, followed by details of the top-up amount in Malaysian Ringgit (RM). The notification consists of the exact date and time of the transaction to complete it. The new account balance is also provided to notify users of their fund availability. This organized format provides readability, ensuring users can monitor their financial activities effectively.



```

order.txt
File Edit View

ORD2024,ShrimpDumplings,2,10.50,21.00,Dine-in, Status: Accepted
ORD2024,PekingDuck,1,49.99,49.99,Dine-in, Status: Accepted
Total Price: 70.99

ORD3033,BeefPho,1,16.99,16.99,Delivery,Status: In Progress
ORD3033,SpringRolls,1,7.25,7.25,Delivery,Status: In Progress
Total Price: 24.24

ORD4044,ChickenKatsu,1,18.50,18.50,Takeaway, Status: Accepted
ORD4044,Onigiri,2,4.75,9.50,Takeaway, Status: Accepted
Total Price: 28.00

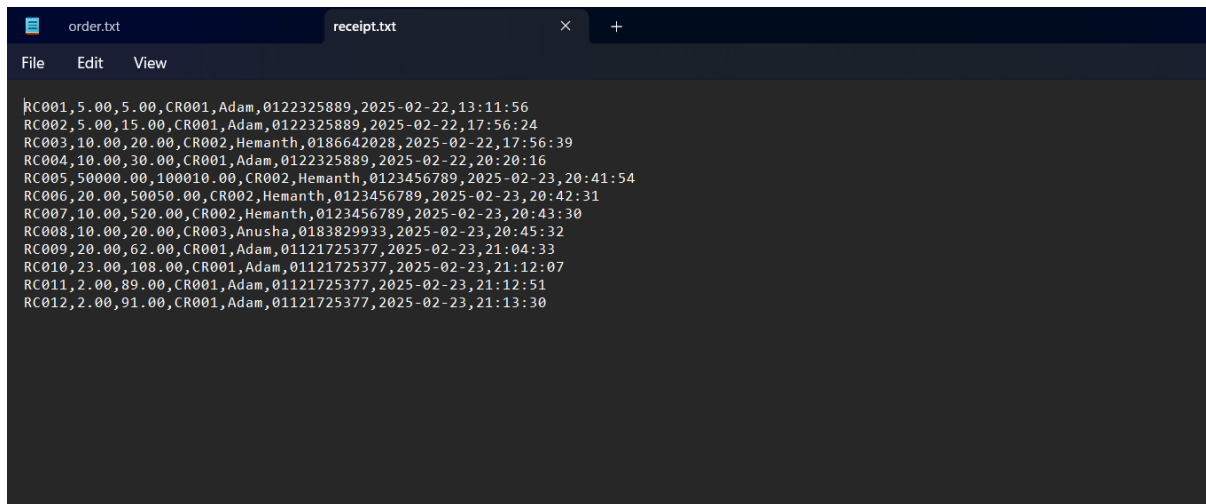
ORD5055,MangoStickyRice,1,8.99,8.99,Dine-in, Status: Cancelled
ORD5055,ThaiGreenCurry,1,15.99,15.99,Dine-in, Status: Cancelled
Total Price: 24.98

ORD6066,GeneralTsosChicken,1,13.50,13.50,Delivery, Status: Cancelled
ORD6066,MontonSoup,1,8.25,8.25,Delivery, Status: Cancelled
Total Price: 21.75

ORD7077,KimchiFriedRice,1,12.75,12.75,Takeaway, Status: Accepted
ORD7077,Soju,1,10.00,10.00,Takeaway, Status: Accepted
Total Price: 22.75
    
```

Figure 58 Order.txt

The order.txt record widely logs food orders with every item's attribute for observation and processing. The record consists of a distinct Order ID, then the item name, quantity, unit price, and total price in Ringgit Malaysia (RM). The order type indicates the type of meal to be eaten in Dine-in, Delivery, or Takeaway format, as an assurance of the clarity of satisfaction. Also, the order status indicates whether the order is Accepted, In Progress, or Cancelled, which indicates its status. A Total Price field at the end of grouped orders provides a total cost breakdown per transaction. The structured format allows for efficient order management, easy tracking, and processing.

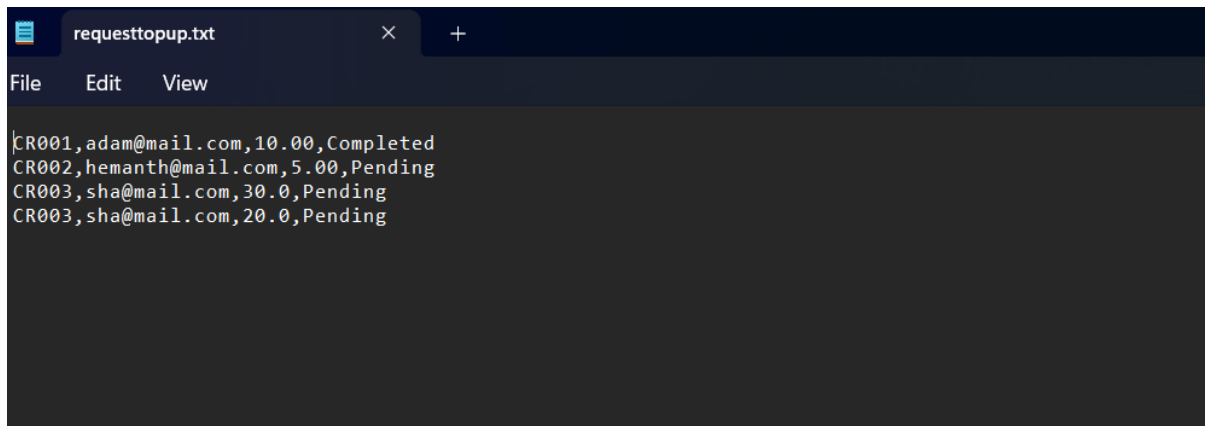


```

RC001,5.00,5.00,CR001,Adam,0122325889,2025-02-22,13:11:56
RC002,5.00,15.00,CR001,Adam,0122325889,2025-02-22,17:56:24
RC003,10.00,20.00,CR002,Hemanth,0186642028,2025-02-22,17:56:39
RC004,10.00,30.00,CR001,Adam,0122325889,2025-02-22,20:20:16
RC005,50000.00,100010.00,CR002,Hemanth,0123456789,2025-02-23,20:41:54
RC006,20.00,50050.00,CR002,Hemanth,0123456789,2025-02-23,20:42:31
RC007,10.00,520.00,CR002,Hemanth,0123456789,2025-02-23,20:43:30
RC008,10.00,20.00,CR003,Anusha,0183829933,2025-02-23,20:45:32
RC009,20.00,62.00,CR001,Adam,01121725377,2025-02-23,21:04:33
RC010,23.00,108.00,CR001,Adam,01121725377,2025-02-23,21:12:07
RC011,2.00,89.00,CR001,Adam,01121725377,2025-02-23,21:12:51
RC012,2.00,91.00,CR001,Adam,01121725377,2025-02-23,21:13:30
    
```

Figure 59 Receipt.txt

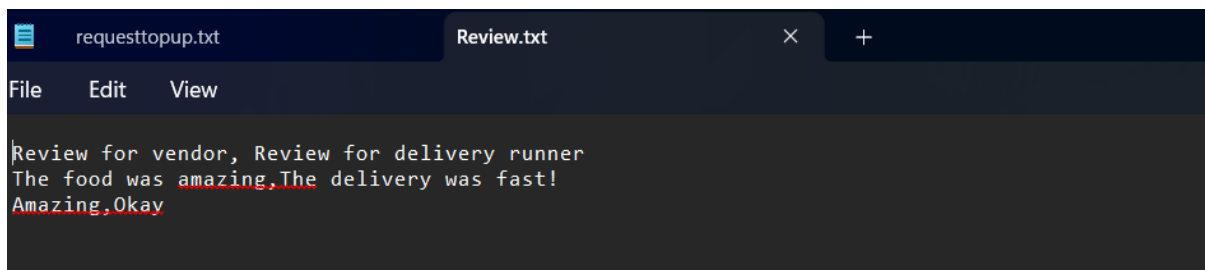
The receipt.txt thoroughly notes transaction information appropriate to customer payments and recharging. From a Receipt ID, each record is then headed by the amount paid in RM, so assisting proper office documentation. The document other holds a customer's unique ID (CRxxxx), customer name, and contact number, thus assisting relationship of each transaction to the related entity. Likewise, the timestamp at the bottom of every entry logs the precise date and time of the transaction, thereby maintaining a proper chronological sequence. This organized structure enables effective management of finances and quick access to previous transactions.



```
requesttopup.txt
File Edit View
CR001,adam@mail.com,10.00,Completed
CR002,hemanth@mail.com,5.00,Pending
CR003,sha@mail.com,30.0,Pending
CR003,sha@mail.com,20.0,Pending
```

Figure 60 RequestTopUp.txt

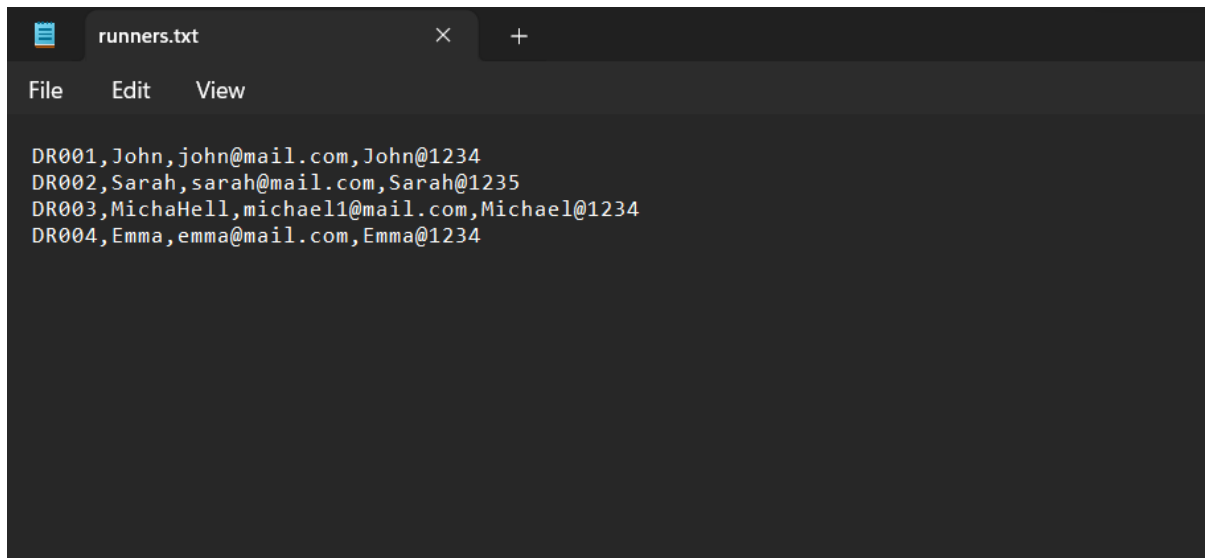
The requesttopup.txt keeps a record of customer top-up requests in a ordered format. Each entry has a Customer ID (CRxxx), the email address, the amount of top-up, and the status of the request. The status possibly will either one be Completed (where the top-up is successful) or Pending (for processing). This is to ease easy tracking of customer top-up transactions so that it will be simple and orderly regarding processing monetary requests.



```
requesttopup.txt Review.txt
File Edit View
Review for vendor, Review for delivery runner
The food was amazing,The delivery was fast!
Amazing,Okay
```

Figure 61 Review.txt

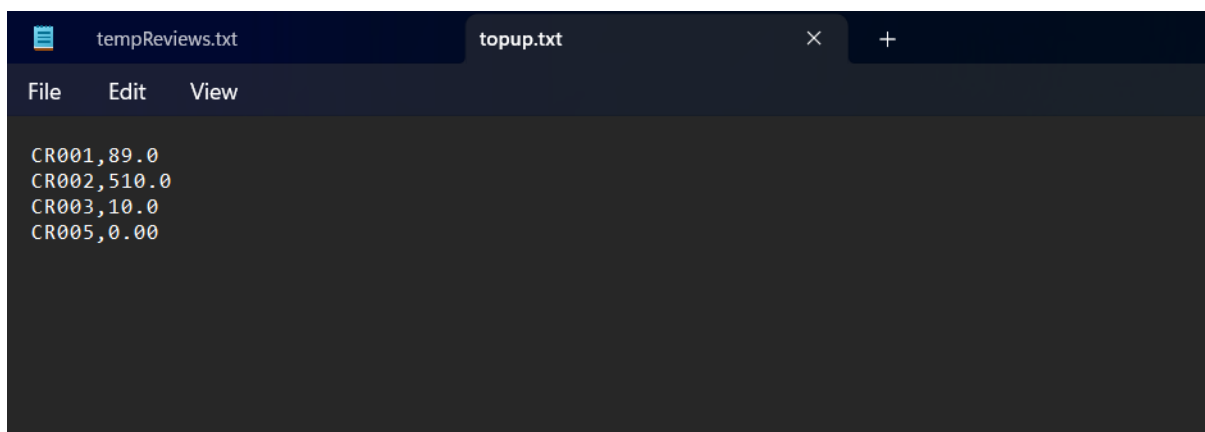
The requesttopup.txt file declares an arranged record of customer top-up requests, ensuring effective tracking and management. Each entry involves of a unique Customer ID (CRxxx), the customer's email address, the top-up amount in RM, and the request status. The status can be either **Completed**, signaling that the top-up has been successfully processed, or Pending, indicating the application is still nearing activity. This organized format allows for seamless monitoring of financial transactions, making it easier to track customer requests and ensure timely processing.



```
runners.txt
File Edit View
DR001,John,john@mail.com,John@1234
DR002,Sarah,sarah@mail.com,Sarah@1235
DR003,Michael,michael1@mail.com,Michael@1234
DR004,Emma,emma@mail.com,Emma@1234
```

Figure 62 runners.txt

The delivery runner details are held in the runners.txt file in a structured format. Each record holds a Runner ID (DRxxx), runner name, email, and system password. The structured format can efficiently manage delivery staff by holding vital information in a well-organized way for easy access. Using a structured record, the system will be able to track and authenticate runners effectively, allowing delivery activities to run smoothly.

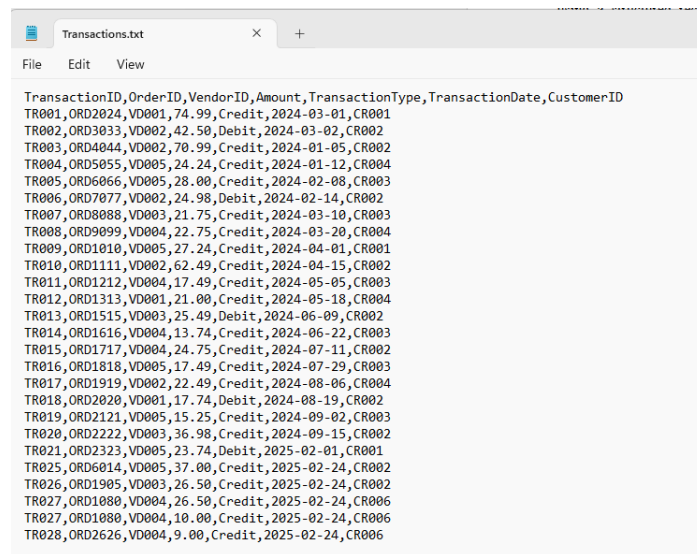


```
tempReviews.txt topup.txt
File Edit View
CR001,89.0
CR002,510.0
CR003,10.0
CR005,0.00
```

Figure 63 Topup.txt

The information for delivery runners are kept in the runners.txt file in a ordered format. A record consists of a Runner ID (DRxxx), the name of the runner, email, and system password. The utilize of a organized format agrees for competent management of delivery staff by filing critical details in an organized form, in this manner improving accessibility. By using a

structured record, the system is capable of tracking and authenticating runners effectively, consequently ensuring efficient execution of delivery tasks.



```

TransactionID,OrderID,VendorID,Amount,TransactionType,TransactionDate,CustomerID
TR001,ORD2024,VD001,74.99,Credit,2024-03-01,CR001
TR002,ORD3033,VD002,42.50,Debit,2024-03-02,CR002
TR003,ORD4044,VD002,70.99,Credit,2024-01-05,CR002
TR004,ORD5055,VD005,24.24,Credit,2024-01-12,CR004
TR005,ORD6066,VD005,28.00,Credit,2024-02-08,CR003
TR006,ORD7077,VD002,24.98,Debit,2024-02-14,CR002
TR007,ORD8088,VD003,21.75,Credit,2024-03-10,CR003
TR008,ORD9099,VD004,22.75,Credit,2024-03-20,CR004
TR009,ORD1010,VD005,27.24,Credit,2024-04-01,CR001
TR010,ORD1111,VD002,62.49,Credit,2024-04-15,CR002
TR011,ORD1212,VD004,17.49,Credit,2024-05-05,CR003
TR012,ORD1313,VD001,21.00,Credit,2024-05-18,CR004
TR013,ORD1515,VD003,25.49,Debit,2024-06-09,CR002
TR014,ORD1616,VD004,13.74,Credit,2024-06-22,CR003
TR015,ORD1717,VD004,24.75,Credit,2024-07-11,CR002
TR016,ORD1818,VD005,17.49,Credit,2024-07-29,CR003
TR017,ORD1919,VD002,22.49,Credit,2024-08-06,CR004
TR018,ORD2020,VD001,17.74,Debit,2024-08-19,CR002
TR019,ORD2121,VD005,15.25,Credit,2024-09-02,CR003
TR020,ORD2222,VD003,36.98,Credit,2024-09-15,CR002
TR021,ORD2323,VD005,23.74,Debit,2025-02-01,CR001
TR025,ORD6014,VD005,37.00,Credit,2025-02-24,CR002
TR026,ORD1905,VD003,26.50,Credit,2025-02-24,CR002
TR027,ORD1080,VD004,26.50,Credit,2025-02-24,CR006
TR027,ORD1080,VD004,10.00,Credit,2025-02-24,CR006
TR028,ORD2626,VD004,9.00,Credit,2025-02-24,CR006

```

Figure 64 Transaction.txt

The Transactions.txt file is a structure of vendor financial transactions. It includes a note of financial transactions in the way of an entry that includes a Transaction ID (TRxxx), the Vendor ID (VDxxx), the Order ID, the transaction amount, the transaction type (either Debit or Credit), the transaction date and the Customer ID (CRxxx). Debit transactions imitate payments withdrawn, and credit transactions means payments received. This format aids easy monitoring of vendor financial transactions, thus enabling the logging and tracing of finances accurately.

6.0 Workload Matrix

Name	Components	Percentage Distribution
HEMANTHRAJAH A/L N KRISHNARAJAH	Manager Delivery runner Compiling	21
ADAM JEROME JOSEPH	Admin	20
ANUSHA DEVY A/P BRAHMATARAN	Customer Object-oriented concepts Customer (Additional feature) Conclusion	21
HEYYMANATHAPILLAY A/L VARATHA RAJU	Delivery Runner Class Diagram Appendix	18
SAILESS RAJ A/L RAJENDRAN	Vendor Use Case Diagram Appendix Limitation Video editing	20